Local Limits of Random Graphs



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Abstract

In this project, we explore the notion of local limits in random graphs. We first define and study essential properties about branching processes, that will serve as approximating objects for the local limit. We will continue by formalising the local limit as the Benjamini-Schramm limit, and give two convenient criteria in terms of the proportion of vertices in the graph. We then present the Configuration Model in full generality, before stating and proving its local limit to a unimodular branching process in the case where the degrees are 1 or 3. We finally end on a new study of the giant component using local weak convergence as an assumption and present an application to the local behaviour of the giant for an Erdős-Rényi random graph.

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Chapter 1

Introduction - On Random Graphs and the Benjamini-Schramm Convergence

Random graphs are central objects in Graph Theory, developed first in 1960 by Paul Erdős and Alfréd Rényi in their paper "On the evolution of random graphs" [1]: Using the same notation, they consider the set $E_{n,N}$ to be the collection of all linear graphs (graphs with no orientation, no parallel edges and no self-loops) of size n, where n is the number of vertices and N the number of edges. A graph in this set is obtained by choosing N out of the possible $M=\binom{n}{2}$ edges. This implies that $|E_{n,N}|=\binom{M}{N}$. Now, a random graph $\Gamma_{n,N}$ is an element of $E_{n,N}$ chosen at random, i.e. with probability $1/|E_{n,N}|$. Other models have been introduced which are similar to the one by the two Hungarian mathematicians. As an important example, the Gilbert's Model named after the American mathematician Edgar Nelson Gilbert, where in a paper written in 1960 [2] he studies graphs in which every edge occurs independently with a probability $p \in (0,1)$. Hence, the probability of obtaining one particular graph with N edges is $p^N(1-p)^{M-N}$, where we still have $M=\binom{n}{2}$. Although constructing graphs seems relatively straightforward, analysing them tends to be rather complex and arduous especially when we consider large random graphs. For this reason, mathematicians have advanced the study of random graphs by simplifying their structure, and the local limit is the answer to the problem.

The notion of local limit has first been described in 2001 by Itai Benjamini and Oded Schramm in [3], and independently in the same time period by Aldous and Steel in [4]. As a result of their work, the *Benjamini-Schramm convergence* (or simply known as the *local limit*) has been extensively used to describe finite sparse random graphs and their properties.
As a key concept in random graph theory, convergence to a limiting tree often occurs as a local limit. Such graphs are said to have a locally tree-like structure and are often described in terms of branching processes. Models such as the Erdős-Rényi random graph, the Configuration Model or the Preferential Attachment Model can all be studied using a local limit to such a process. As an even more profound role, "if a model has a local weak limit, then any local property converges to a specific measure on the limit graph. Hence, if we know the local graph limit, we can obtain the limits of a wide variety of local graph measures" [5, pp.2]. Considering some interesting examples, knowing the local limit of a graph, we can investigate the convergence of the neighbourhoods and even the convergence of its

clustering coefficient. As a consequence, we can also sometimes make conclusions about global properties of the graph where they can also be studied using the local behaviour of the random graph. It is specifically the case for the study of the greatest component in a graph, known as the giant component.

In Chapter 2, we describe the limiting graphs with the notion of branching processes. We define important properties such as the survival probability, the phase transitions and the total progeny for general processes. We end the chapter with two important examples that will be used in the later sections, comprising of a Poisson branching process and a Unimodular branching process. In Chapter 3, we formally define the notion of local limits by first presenting the metric space of rooted graphs before stating the definitions on local weak convergence in distribution and in probability. We then end this section on the presentation of two criteria for local convergence, that are generally used as easier methods for proving local limits of particular random graphs. In the fourth chapter we study the Configuration Model in full generality along with a discussion on the simplicity of the model, before moving on to its local limit where we prove that the configuration model locally converges to a unimodular branching process in the simplest non-trivial case, where the degrees are either 1 or 3. We finally end this thesis on a fifth chapter where we investigate the giant component and how the local limit can describe its structure. We close the section with an application of the general theory to an Erdős-Rényi Random Graph $ER_n(\frac{\lambda}{n})$.

Chapter 2

Branching Processes

Branching processes are an important notion for the study of random graphs as they can describe some of their connected components. We will describe branching processes in a general setup before moving on to the study of branching processes with a Poisson offspring distribution and unimodular branching processes. Results and proofs in this chapter are mainly from the first volume of Remco van der Hofstad's book on "Random Graphs and Complex Networks" [6].

2.1 Definition and General Survival

A (Galton-Watson) branching process is the simplest possible model for a population evolving in time, where we have a relation between the birth of individuals with equal probability and the offspring distribution given by the probability that an individual has a certain number of children.
Let's first define some terminology. In this section, we follow the notation given in [6] and we denote by $(p_i)_{i\geq 0}$ the offspring distribution, where:

$$p_i = \mathbb{P}[\text{individual has } i \text{ children}]. \tag{2.1}$$

Now considering $(X_{n,i})_{n,i\geq 0}$ to be a doubly infinite array of i.i.d. random variables, we denote by Z_n the number of individuals in the $n'th$ generation, where $Z_0 = 1$, and satisfies the following relation:

$$Z_n = \sum_{i=1}^{Z_{n-1}} X_{n,i}. \tag{2.2}$$

Note that we will also write X as the offspring distribution, such that $X_{n,i} \sim X$ for all $n, i \geq 0$.
Finally, we define the *extinction probability* η by:

$$\eta = \mathbb{P}[\exists n : Z_n = 0], \tag{2.3}$$

and naturally the *survival probability* $\chi = 1 - \eta$ by:

$$\chi = \mathbb{P}[Z_n > 0 \ \ \forall n \geq 0]. \tag{2.4}$$

We can now examine the conditions for a branching process to survive by studying its phase transitions, after stating an important theorem about the survival probability.

Theorem 2.1.1. *The survival probability χ is the smallest solution in $[0,1]$ of*

$$\chi = 1 - G_X(\eta), \tag{2.5}$$

where $s \mapsto G_X(s)$ is the probability generating function of the offspring distribution X, i.e.,

$$G_X(s) = \mathbb{E}[s^X]. \tag{2.6}$$

Proof. For simplicity, we will prove these results in terms of the *extinction probability* η first; based on the proof given in [6].

• Let $\eta_n = \mathbb{P}(Z_n = 0)$. Since $\{Z_n = 0\} \subseteq \{Z_{n+1} = 0\}$ then we know that η_n is an increasing sequence bounded above by 1. Hence, η_n has a limit $\eta \in [0,1]$.
Consider now $G_n(s) = \mathbb{E}[s^{Z_n}]$ to be the probability generating function of the nth generation.
Using the Law of Iterated Expectations and conditioning on the first generation, we have:

$$G_n(s) = \mathbb{E}[s^{Z_n}] = \sum_{i=0}^{\infty} \mathbb{P}(Z_1 = i)\mathbb{E}(s^{Z_n}|Z_1 = i) = \sum_{i=0}^{\infty} p_i \mathbb{E}(s^{Z_n}|Z_1 = i). \tag{2.7}$$

Now, we can consider $(Y_{n-1,j})_{j=1}^{Z_1}$ to be the branching process of the jth individual of the first generation. Then it is clear that $Z_n = \sum_{j=1}^{Z_1} Y_{n-1,j}$, using the convention that the empty sum in the case $i = 0$ is equal to zero. Therefore:

$$G_n(s) = \sum_{i=0}^{\infty} p_i \mathbb{E}(s^{\sum_{j=1}^{Z_1} Y_{n-1,j}}|Z_1 = i) = \sum_{i=0}^{\infty} p_i \prod_{j=1}^{i} \mathbb{E}(s^{Y_{n-1,j}}) = \sum_{i=0}^{\infty} p_i [\mathbb{E}(s^{Y_{n-1,1}})]^i = \sum_{i=0}^{\infty} p_i \left(G_{n-1}(s)\right)^i, \tag{2.8}$$

where we have used the fact that each of the i individuals produces offspring in the nth generation in an i.i.d. way, with probability generating function $s \mapsto G_{n-1}(s)$.
Hence we conclude that $G_n(s) = G_1(G_{n-1}(s))$, since $G_1(s) = \mathbb{E}(s^{Z_1}) = \sum_{i=1}^{\infty} p_i s^i = G_X(s)$.
Setting $s = 0$ we obtain $\eta_n = G_X(\eta_{n-1})$. And since G_X is continuous in $[0,1]$ we have that:

$$\lim_{n\to\infty} \eta_n = \lim_{n\to\infty} G_X(\eta_{n-1}) = G_X(\lim_{n\to\infty} \eta_{n-1}), \tag{2.9}$$

which implies that $\eta = G_X(\eta) \Rightarrow \chi = 1 - G_X(\eta)$ as required.

• Let's now show that χ is the smallest solution of equation (2.5).
Suppose that $\psi \in [0,1]$ satisfies $\psi = G_X(\psi)$. We claim that $\eta \leq \psi$. Let's show by induction that $\eta_n \leq \psi$ for all n.
For $n = 0$, $\eta_0 = 0 \leq \psi$ which initialises the induction.
Suppose now that $\eta_n \leq \psi$ for the first $(n-1)$ terms. Let's show that this still holds for the term n.

Recall that $\eta_n = G_X(\eta_{n-1})$ and that $G_X(s)$ is increasing on $[0,1]$. Then:

$$\eta_n = G_X(\eta_{n-1}) \leq G_X(\psi) = \psi. \tag{2.10}$$

Since $\eta_n \to \eta$ we have that $\eta \leq \psi$ as required. Therefore η is the smallest solution of the equation $\eta = G_X(\eta)$ which implies that χ is the smallest solution of equation (2.5).

□

Theorem 2.1.2 (Phase transitions for branching processes)**.** *Consider a branching process with i.i.d. offspring X. Then:*

$$\begin{aligned}
&(i)\ \ \chi = 0 \ \text{ when } \ \mathbb{E}(X) < 1 \quad \textit{(Subcritical regime)},\\
&(ii)\ \ \chi > 0 \ \text{ when } \ \mathbb{E}(X) > 1 \quad \textit{(Supercritical regime)},\\
&(iii)\ \ \chi = 0 \ \text{ when } \ \mathbb{E}(X) = 1,\ \ \mathbb{P}(X=1) < 1 \quad \textit{(Critical regime)}.
\end{aligned}$$

Proof. For simplicity, we will again prove the results in terms of the *extinction probability* η first. First note that if $\mathbb{P}(X=1) = 1$ then $\mathbb{E}(X) = 1$ and for all n, $Z_n = 1 \Rightarrow \eta = 0$. Furthermore, if $\mathbb{P}(X \leq 1) = 1$ and $p^* = \mathbb{P}(X=0) > 0$ then:

$$\mathbb{P}(Z_n = 0) = 1 - \mathbb{P}(Z_n = 1) = 1 - [\mathbb{P}(Z_1 = 1)]^n = 1 - (1-p^*)^n \xrightarrow{n\to\infty} 1. \tag{2.11}$$

Now we are left to consider the case where $\mathbb{P}(X \leq 1) < 1$. Then in this case and by the properties of probability generating functions, $G_X(s)$ is strictly increasing and strictly convex on $[0,1]$.
Assume $G_X(0) > 0$.
(i) When $\mathbb{E}(X) < 1$ there is a unique solution which is $\eta = 1$. It implies that $\chi = 0$.
(ii) When $\mathbb{E}(X) > 1$ there exists two distinct solutions which are η and 1. Hence $\eta < 1 \Rightarrow \chi > 0$.
(iii) When $\mathbb{E}(X) = 1$ there exists a unique solution which is $\eta = 1$. It implies that $\chi = 0$.

□

2.2 Total Progeny and Mean Family Size

We now define the total progeny of the branching process T as the total number of individuals up through generation n. More formally we have that:

$$T = \sum_{n=0}^{\infty} Z_n. \tag{2.12}$$

We can now present two theorems on the first moments of Z_n and T.

Theorem 2.2.1 (First moment of Z_n)**.** *For all $n \geq 0$, and with $\mu = \mathbb{E}(Z_1) = \mathbb{E}(X)$ the expected offspring of a given individual,*

$$\mathbb{E}(Z_n) = \mu^n. \tag{2.13}$$

Proof. Recall equation (2.2) where $X_{n,i}$ is a doubly infinite array of i.i.d. random variables. We note that $(X_{n,i})_{i\geq 1}$ is independent of Z_{n-1}. Let's prove this theorem by induction on n.
For $n=0$, $\mathbb{E}(Z_0)=1=\mu^0$ which initializes the induction.
Suppose that $\mathbb{E}(Z_n)=\mu^n$ for the first $(n-1)$ terms. Let's show that this still holds for the term n.
Using the Law of Iterated Expectations:

$$\mathbb{E}(Z_n)=\mathbb{E}\big(\mathbb{E}(Z_n \mid Z_{n-1})\big)=\mathbb{E}\left[\mathbb{E}\left(\sum_{i=1}^{Z_{n-1}} X_{n,i}\middle| Z_{n-1}\right)\right]=\mathbb{E}(Z_{n-1}\mu|Z_{n-1})=\mu\mathbb{E}(Z_{n-1})=\mu^n. \quad (2.14)$$

□

Theorem 2.2.2 (Expected total progeny)**.** *For a branching process with i.i.d. offspring X having mean offspring $\mu<1$,*

$$\mathbb{E}(T)=\frac{1}{1-\mu}. \quad (2.15)$$

Proof. We have,

$$\mathbb{E}(T)=\mathbb{E}\left[\sum_{n=0}^{\infty} Z_n\right]=\sum_{n=0}^{\infty}\mathbb{E}(Z_n)=\sum_{n=0}^{\infty}\mu^n=\frac{1}{1-\mu}, \quad (2.16)$$

where we have used the linearity of the expectation, the fact that $\mu<1$ and Theorem 2.2.1. □

We will now end this section on an important theorem investigating the so-called *Law of Total Progeny*. We will omit the proof of this theorem as it would take us beyond the scope of this paper, as it involves Random Walks and the *Hitting-time Theorem*; where details are left to the interest of the reader and can be found in [6, pp.98].

Theorem 2.2.3 (Law of Total Progeny)**.** *For a branching process with i.i.d. offspring distribution $Z_1=X$, where $(X_i)_{i=1}^n$ are i.i.d. copies of X,*

$$\mathbb{P}(T=n)=\frac{1}{n}\mathbb{P}\left(\sum_{k=1}^{n} X_k=n-1\right). \quad (2.17)$$

2.3 Two Important Branching Processes

We now end this chapter by studying two particular types of processes, which are Poisson branching processes and unimodular branching processes.

2.3.1 Poisson Branching Processes

We first start by presenting Poisson branching processes. They are branching processes with a Poisson offspring distribution. They often arise in the theory of random graphs as they can describe a large variety of properties, including the famous convergence in probability of the Erdős-Rényi Random Graph to a Poisson branching process. For the remainder of this paper we denote the distribution of a Poisson branching process by $\mathbb{P}_\lambda$, the total progeny of the Poisson branching process by T_λ and the Poisson random variable by X_λ. Note also that, whenever a λ appears as a subscript, it is to remark the

explicit dependence of a given quantity on λ. We can first derive the probability generating function and the survival probability for a Poisson branching process. We have:

$$G_\lambda(s) = \mathbb{E}_\lambda(s^{X_\lambda}) = \sum_{i=0}^{\infty} s^i e^{-\lambda} \frac{\lambda^i}{i!} = e^{-\lambda} \sum_{i=0}^{\infty} \frac{(s\lambda)^i}{i!} = e^{-\lambda} e^{s\lambda} = e^{\lambda(s-1)}, \tag{2.18}$$

and using equation (2.5) we have:

$$\chi_\lambda = 1 - e^{-\chi_\lambda}. \tag{2.19}$$

We now derive the total progeny for a Poisson branching process.

Theorem 2.3.1 (Total progeny for a Poisson branching process)**.** *For a branching process with i.i.d. offspring X_λ following a Poisson distribution with mean λ, and with $n \geq 1$,*

$$\mathbb{P}_\lambda(T_\lambda = n) = \frac{(\lambda n)^{n-1}}{n!} e^{-\lambda n}. \tag{2.20}$$

In fact, we can observe that the total progeny for a Poisson branching process follows the Borel distribution with parameter λ.

Proof. We use the Law of Total Progeny (Theorem 2.2.3) where,

$$\begin{aligned}
\mathbb{P}_\lambda(T_\lambda = n) = \frac{1}{n}\mathbb{P}\left(\sum_{k=1}^{n} (X_\lambda)_k = n-1\right) &= \frac{1}{n}\mathbb{P}\big(n(X_\lambda)_1 = n-1\big) \\
&= \frac{1}{n}(\lambda n)^{n-1} \frac{1}{(n-1)!} e^{-\lambda n} \\
&= \frac{(\lambda n)^{n-1}}{n!} e^{-\lambda n},
\end{aligned}$$

where we have used the fact: if $X_i \sim Poi(\lambda)$ (i.i.d. random variables) then $\sum_{i=1}^{n} X_i = nX_1 \sim Poi(n\lambda)$. □

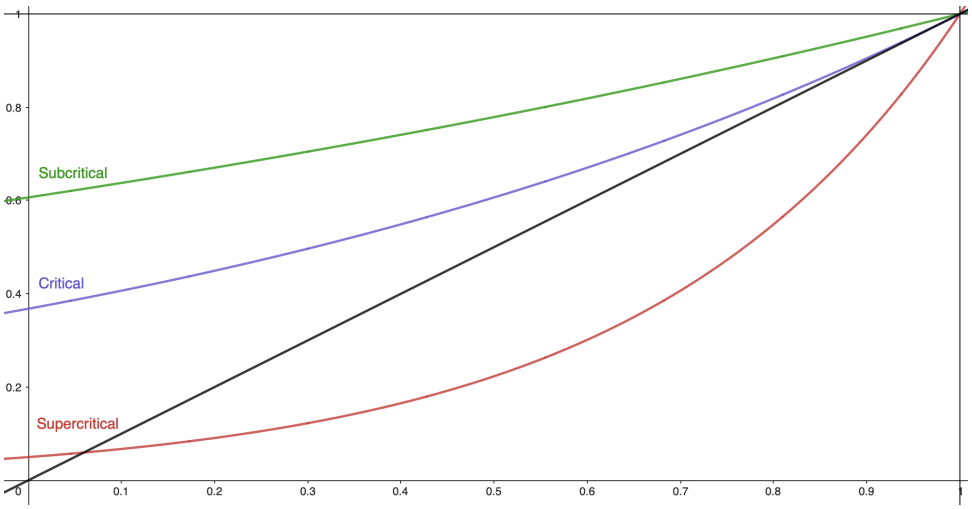


Figure 2.1: Plot of the phase transitions for a Poisson branching process for different values of $\mathbb{E}(X)$. [This figure has been created using GeoGebra(2024)]

2.3.2 Unimodular Branching Processes

Our second example is a unimodular branching process. This type of process will later be useful to describe the local limit of the configuration model. We first define the notion of size-biased distribution.

Definition 2.3.2. *Consider a non-negative random variable X. Then, X^* is called the size-biased version of X, where its distribution function F_{X^*} is given by:*

$$F_{X^*} := \mathbb{P}(X^* \leq x) = \frac{\mathbb{E}\left[X \mathbb{1}_{\{X \leq x\}}\right]}{\mathbb{E}[X]}. \tag{2.21}$$

Now, in this process, we fix a probability distribution $(p_k)_{k\geq 1}$ corresponding to a degree distribution. Then we have that $p_k = \mathbb{P}(D = k)$ for some integer random variable D. A branching process is said to be *unimodular* with root offspring distribution $(p_k)_{k\geq 1}$ if the root has offspring distribution $(p_k)_{k\geq 1}$ while all other vertices have root offspring distribution $(p^*_k)_{k\geq 1}$, where it is given by:

$$p^*_k = \mathbb{P}(D^* - 1 = k) = \frac{(k+1)}{\sum_{i\geq 0} i p_i} p_{k+1} = \frac{(k+1)}{\mathbb{E}(D)} p_{k+1}. \tag{2.22}$$

In the above, D^* is the size-biased version of D as defined in Definition 2.3.2; and call p^*_k the size-biased degree distribution.

As explained in a paper on Unimodular random trees by Itai Benjamini, Russell Lyons, and Oded Schramm [7], the idea behind unimodularity arises in some particular cases of probability measures in finite graphs. "Intuitively, a probability measure on rooted graphs is unimodular if and only if its root is chosen uniformly at random among all its vertices" [7]. One might also be interested to define this notion on infinite graphs, however we would need to formally define the notion of the Mass-Transport Principle for such a study, which would take us beyond the scope of this paper.

Chapter 3

Local Limits

We now start the main chapter of this paper. We will qualitatively describe the notion of local limits by first introducing the metric space of rooted graphs before moving on to the definition of the local weak convergence in the setting of random graphs along with two criteria for local convergence. In this chapter, the use of rooted graphs will be essential since we will study the local limit in the neighbourhood of a random graph, precisely around its root ρ_n.

3.1 The Metric Space of Rooted Graphs

In this first section, we will briefly but formally describe the metric space of rooted graphs. Such a space is necessary to give a notion of distance on graphs, since local convergence statements are based on such a space. For completeness we will include a brief explanation of the theory but we will not supply a detailed explanation of this metric space. Our goal, however, is to keep in mind that such a space exists and that a metric can be exploited. The following theory will follow closely the second volume of R. van der Hofstad's book on "Random Graphs and Complex Networks" [8]. Further details on the metric space of rooted graphs may be found in the following paper [9].
Throughout this section, we consider two rooted graphs (G_1, ρ_1) and (G_2, ρ_2).

Definition 3.1.1 (Neighbourhoods on rooted graphs)**.** *Consider a rooted graph (G, ρ) and let $B_r^{(G)}(\rho)$ be the subgraph of (G, ρ) such that all its vertices are at distance at most $r \geq 1$ away from the root ρ. In fact, considering the vertex set $V(B_r^{(G)}(\rho))$ and the edge set $E(B_r^{(G)}(\rho))$ of $B_r^{(G)}(\rho)$ we have that:*

$$V(B_r^{(G)}(\rho)) = \{v \in V(G) : dist_G(\rho, v) \leq r\}, \tag{3.1}$$

$$E(B_r^{(G)}(\rho)) = \{\{u, v\} \in E(G) : dist_G(\rho, v) \leq r \text{ and } dist_G(\rho, u) \leq r\}. \tag{3.2}$$

We will now define the notion of isomorphism of graphs assuming that this notion actually makes sense. Informally, this notion is to describe the graphs that are "similar".

Definition 3.1.2 (Isomorphism of graphs)**.** *Two rooted graphs are said to be isomorphic, denoted by $(G_1, \rho_1) \cong (G_2, \rho_2)$, if there exists a bijection $\psi : V(G_1) \longrightarrow V(G_2)$ such that $\psi(\rho_1) = \rho_2$ and for $\{u, v\} \in E(G_1)$ we have $\{\psi(u), \psi(v)\} \in E(G_2)$.*

We now have all the tools to present the metric on rooted graphs.

Definition 3.1.3 (Metric on rooted graphs)**.** *Consider two connected rooted graphs, and denote by* $B_r^{(G_1)}(\rho_1)$, $B_r^{(G_2)}(\rho_2)$ *the neighbourhoods of the vertices* $\rho_1 \in V(G_1)$, $\rho_2 \in V(G_2)$. *Let also*

$$K = \sup\{r : B_r^{(G_1)}(\rho_1) \cong B_r^{(G_2)}(\rho_2)\}. \tag{3.3}$$

We then define the metric on rooted graphs by:

$$d_{\mathscr{G}}\big((G_1,\rho_1),(G_2,\rho_2)\big) = \frac{1}{K+1}, \tag{3.4}$$

where K is the largest value of r for which $B_r^{(G_1)}(\rho_1) \cong B_r^{(G_2)}(\rho_2)$.

Note that $K = \infty$ is allowed and implies that $B_r^{(G_1)}(\rho_1)$ and $B_r^{(G_2)}(\rho_2)$ are isomorphic for all $r \geq 1$. From now on, we write $(\mathscr{G}, d_{\mathscr{G}})$ to be the metric space of rooted graphs under the metric $d_{\mathscr{G}}$.

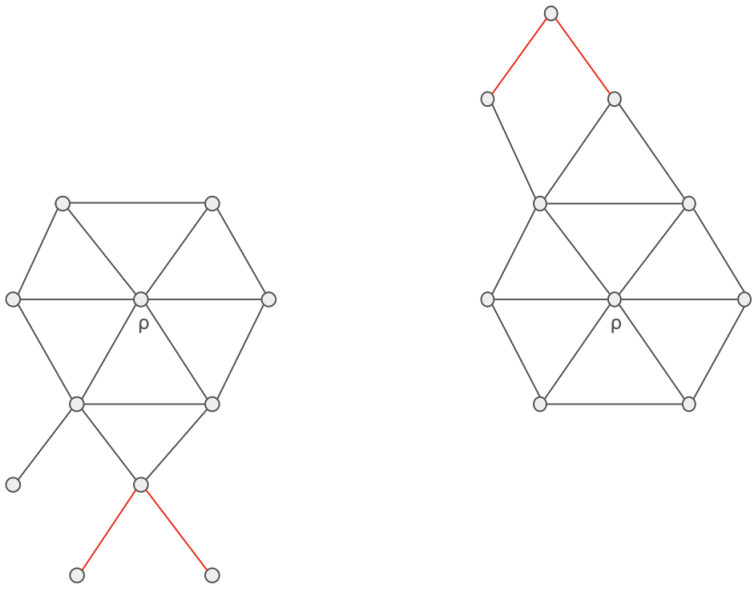


Figure 3.1: Representation of two rooted graphs for which the isomorphism breaks when $r > 2$. In this case, $K = 2$ and $d_{\mathscr{G}}\big((G_1,\rho_1),(G_2,\rho_2)\big) = \frac{1}{3}$. [This figure has been created using GeoGebra(2024)]

3.2 Defining Local Limits

Local limits are an important concept to study possibly large and complex graphs. They give a tool to describe the notion that, in a graph, from the point of view of a vertex chosen at random and naturally creating a rooted graph, the resulting structure looks like a certain limiting graph. In fact, the local limit is sometimes seen as the best role to describe properties of various models. In this section, we will formally define the local limit as the local convergence of random graphs and study some of its resulting properties.

We first start by defining two cases of the local weak convergence, one for the convergence in distribution and the other for the convergence in probability. [10]

Definition 3.2.1 (Local weak convergence in distribution)**.** *Consider a sequence (G_n, ρ_n) of random graphs with random root ρ_n. Let (H, ρ_H) be a rooted graph. Then, we say that (G_n, ρ_n) converges locally weakly in distribution to (G, ρ) if, for all $r \geq 1$, and for all rooted graphs $H \in \mathscr{G}$,*

$$\lim_{n\to\infty} \mathbb{P}\big(B_r^{G_n}(\rho_n) \cong (H, \rho_H)\big) = \mathbb{P}\big(B_r^{G}(\rho) \cong H\big). \tag{3.5}$$

In this case we can write $(G_n, \rho_n) \xrightarrow{d} (G, \rho)$.

Definition 3.2.2 (Local weak convergence in probability)**.** *Consider a sequence (G_n, ρ_n) of random graphs with random root ρ_n. Let (H, ρ_H) be a rooted graph. Then, we say that (G_n, ρ_n) converges locally weakly in probability to (G, ρ) if, for all $r \geq 1$, and for all rooted graphs $H \in \mathscr{G}$,*

$$\mathbb{P}\big(B_r^{G_n}(\rho_n) \cong (H, \rho_H) | G_n\big) \xrightarrow{\mathbb{P}} \mathbb{P}\big(B_r^{G}(\rho) \cong H\big). \tag{3.6}$$

In the settings of the definitions, we can note two sources of randomness. First, and it may seem unsurprising, the random graph plays a central part. Secondly, a less trivial source is the choice for the root in G_n. This fact holds provided that the graph is not vertex-transitive, which means that the choice for the root is not important and will not change the final result, in which case $\mathbb{P}\big(B_r^{G_n}(\rho_n) \cong (H, \rho_H)\big) = \mathbb{P}\big(B_r^{G_n}(\rho_1) \cong (H, \rho_H)\big)$. Assuming it is not vertex-transitive, and it is usually in these instances that local limit results are the most interesting, we may have to deal with roots that are chosen randomly, as for example we can chose the root uniformly at random in the graph.
Now, comparing (3.5) and (3.6), we can observe a key difference in the statements regarding randomness. In Definition 3.2.1, we take a probability over both sources of randomness, and in the limit it leads to a deterministic quantity. However, in Definition 3.2.2, the left hand side is a random variable due to the dependence of the random graph G_n. Note that (3.6) does not depend on the choice of the root ρ_n.

Remark 3.2.3 (Relation between local limits)**.** *One can show that local convergence in probability implies local convergence in distribution. However, the converse is not always true. To motivate this we consider the following example where $p \in (0,1)$. Let $(G_n)_{n\geq 1}$ be an i.i.d. sequence of random graphs where components have size 1 or 2, i.e. either we have an isolated vertex (denoted by the event $\widehat{u}$) or a connected pair of vertices (denoted by the event $\overline{uv}$). Assume that in G_n, $\mathbb{P}(\widehat{u}) = p$ and $\mathbb{P}(\overline{uv}) = 1-p$. Consider also that the limiting graph G has $\mathbb{P}(\widehat{u}) = p$ and $\mathbb{P}(\overline{uv}) = 1-p$. We remark that both graphs follow a form of Bernoulli random variable $Ber(p)$. Let's now study the r-neighbourhood of G_n.*
- If $r = 0$, $\mathbb{P}\big(B_0^{G_n}(\rho_n) \cong (H, \rho_H)\big) \xrightarrow{n\to+\infty} \mathbb{P}(\widehat{u}) = p$.
- If $r = 1$, $\mathbb{P}\big(B_1^{G_n}(\rho_n) \cong (H, \rho_H)\big) \xrightarrow{n\to+\infty} \mathbb{P}(\overline{uv}) = 1-p$.
- If $r \geq 2$, the neighbourhood of G_n will not contain more vertices since components are of size at most 2.
Therefore we have that $(G_n, \rho_n) \xrightarrow{d} (G, \rho)$. However for any $\varepsilon \in (0,1)$, if we now consider relation

(3.6), abbreviating it by $G_n \xrightarrow{\mathbb{P}} G$*, we have that local weak convergence in probability fails since:* $\mathbb{P}(|G_n - G| > \varepsilon) = \mathbb{P}(G_n = \widehat{u}, G = \overline{uv}) + \mathbb{P}(G_n = \overline{uv}, G = \widehat{u}) \xrightarrow{n\to+\infty} 2p(1-p) \neq 0.$

3.3 Criterion for Local Convergence

We now state two criteria for local convergence in distribution and in probability. First, we define the quantity $p^{(G_n)}(H)$ to be the proportion of vertices v for which the isomorphism $B_r^{G_n}(v) \cong H$ holds, and is formally defined by:

$$p^{(G_n)}(H) = \frac{1}{|V(G_n)|} \sum_{v \in V(G_n)} \mathbb{1}_{\{B_r^{G_n}(v) \cong H\}}. \tag{3.7}$$

Criterion 3.3.1. *[8, pp.58]Let* $(G_n)_{n\geq 1}$ *be a sequence of graphs. Then for all rooted graphs* $H \in \mathscr{G}$ *and integers* $r \geq 0$ *if*

$$\mathbb{E}[p^{(G_n)}(H)] := \frac{1}{|V(G_n)|} \sum_{v \in V(G_n)} \mathbb{P}\big(B_r^{G_n}(v) \cong H\big) \xrightarrow{n\to+\infty} \mathbb{P}\big(B_r^{G}(\rho) \cong H\big), \tag{3.8}$$

then, $(G_n, \rho_n) \xrightarrow{d} (G, \rho)$.

Criterion 3.3.2. *[8, pp.58]Let* $(G_n)_{n\geq 1}$ *be a sequence of graphs. Then for all rooted graphs* $H \in \mathscr{G}$ *and integers* $r \geq 0$ *if*

$$p^{(G_n)}(H) \xrightarrow{\mathbb{P}} \mathbb{P}\big(B_r^{G}(\rho) \cong H\big), \tag{3.9}$$

then, G_n *converges locally in probability to* (G, ρ).

One might wonder about the point behind these two statements. In general proofs for the local limit of random graphs, it is generally not convenient to use Definition 3.2.1 and Definition 3.2.2. It is in fact easier to work with Criterion 3.3.1 and Criterion 3.3.2 since they can be coupled to a first or second moment method. For example, after studying several proofs on local convergence in probability, mainly from R. van der Hofstad's book [8], we can summarise the general steps using Criterion 3.3.2 with a second moment method as follows:

1) We calculate the first moment of $p^{(G_n)}(H)$ and show that:

$$\mathbb{E}[p^{(G_n)}(H)] \xrightarrow{n\to+\infty} \mathbb{P}\big(B_r^{G}(\rho) \cong H\big). \tag{3.10}$$

2) We calculate the second moment of $p^{(G_n)}(H)$ and show that it is approximately equal to its first moment squared:

$$\mathbb{E}\big[\big(p^{(G_n)}(H)\big)^2\big] \xrightarrow{n\to+\infty} \big[\mathbb{P}\big(B_r^{G}(\rho) \cong H\big)\big]^2. \tag{3.11}$$

3) From the first two steps, we conclude that:

$$Var\big(p^{(G_n)}(H)\big) \xrightarrow{n\to+\infty} 0 \Longrightarrow Var\big(p^{(G_n)}(H)\big) = o(1). \tag{3.12}$$

4) Finally, using a concentration inequality such as Chebyshev's Inequality we obtain:

$$\mathbb{P}\Big(\big|p^{(G_n)}(H) - \mathbb{P}\big(B_r^G(\rho) \cong H\big)\big| > \varepsilon\Big) \leq \frac{Var\big(p^{(G_n)}(H)\big)}{\varepsilon^2} = \frac{o(1)}{\varepsilon^2}. \tag{3.13}$$

Hence for every $\varepsilon > 0$,

$$\lim_{n\to+\infty} \mathbb{P}\Big(\big|p^{(G_n)}(H) - \mathbb{P}\big(B_r^G(\rho) \cong H\big)\big| > \varepsilon\Big) = 0 \Longrightarrow p^{(G_n)}(H) \xrightarrow{\mathbb{P}} \mathbb{P}\big(B_r^G(\rho) \cong H\big). \tag{3.14}$$

Overall, this proves local convergence in probability.

In the next section, we will prove the local limit of the Configuration Model with a constrained degree sequence. However we will not use the method described above, but we will rather use a combinatorial approach to the problem. Nevertheless, in general literature, the second moment method is always preferred in more advanced and global setups.

Chapter 4

The Configuration Model

We are now going to shift our interest to the main example of this paper: the Configuration Model (CM). The aim of this chapter is to prove that, when vertices have degree 1 or 3, the local limit of the Configuration Model is a unimodular branching process. We will first introduce the model in its full generality, engaging in discussions about the simplicity of the model and particular properties, before applying this theory to the constrained degree sequence.
We will in general follow a paper written by Omer Angel, Remco van der Hofstad and Cecilia Holmgren [11] along with [8] for the main parts of this chapter.

4.1 Introduction to the Configuration Model

In this first section, we will present the Configuration Model, including its parameters and conditions. We will then describe how to construct the random graph, and set up some notation for the remainder of the chapter.

4.1.1 Presentation and Construction of the Model

The Configuration Model is a random graph with a prescribed degree sequence $\mathbf{d} = (d_1, \dots, d_n)$, called a configuration, where vertex 1 has degree d_1, vertex 2 has degree d_2, et cetera. We will, from now on, refer to $CM_n(\mathbf{d})$ for the model.
Given this setup, we would like to construct such a random graph with prescribed degree sequence $\mathbf{d}$. For this, we consider l_n half-edges satisfying $l_n = \sum_{i \in [n]} d_i$, where $[n] := \{1, 2, \dots, n\}$. In this model, the sum of the degrees l_n is taken to be even. Algorithmically, we can construct the CM as follows: take l_n half-edges, and start a random process of matching them. In particular, we randomly choose two half-edges and connect them to form an edge, and remove the two chosen half-edges from future matchings. We continue this process until all half-edges have been paired and end up with a particular configuration of $CM_n(\mathbf{d})$ with degree sequence $\mathbf{d}$.
Note that this process justifies why we have to consider an even number of half-edges, otherwise we would be left with one unpaired half-edge, which would not end the algorithm to create a particular configuration.

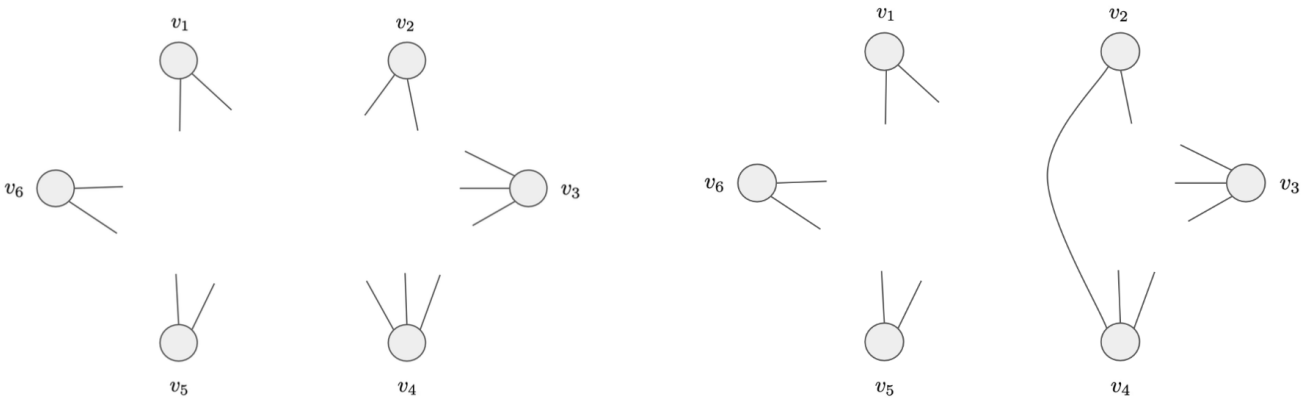


Figure 4.1: First steps in the construction of $CM_n(\mathbf{d})$ using a random matching of half-edges in the case $n = 6$ and $l_n = 14$. [This figure has been created using GeoGebra(2024)]

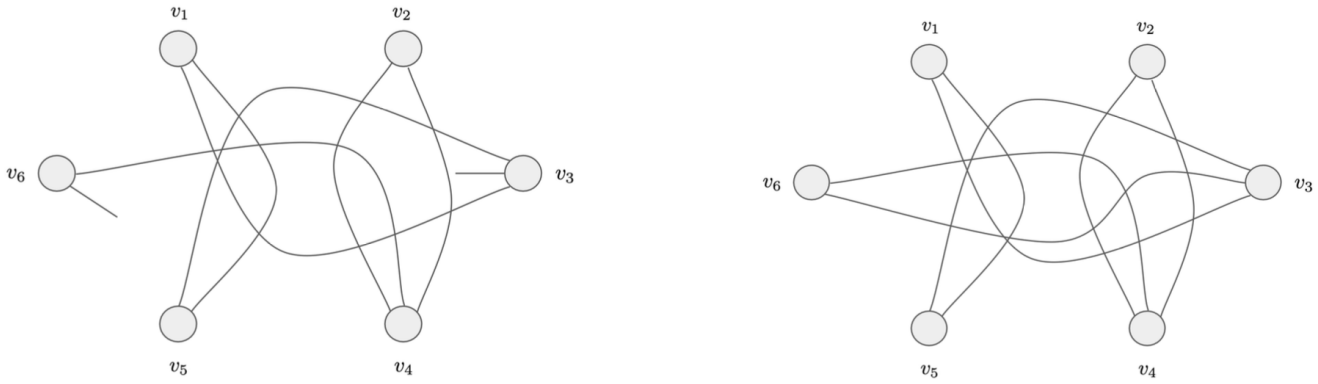


Figure 4.2: Final steps in the construction of $CM_n(\mathbf{d})$ using a random matching of half-edges in the case $n = 6$ and $l_n = 14$. The final random graph is a particular configuration of the CM. [This figure has been created using GeoGebra(2024)]

4.1.2 Properties of $CM_n(\mathbf{d})$

Now that we know how the random graph is being constructed, we can state some basic and more evolved properties about $CM_n(\mathbf{d})$.

Theorem 4.1.1 (Number of matchings)**.** *In $CM_n(\boldsymbol{d})$, there are $(l_n-1)!! = (l_n-1)\times(l_n-3)\times\ldots\times 3\times 1$ possible matchings, among the original l_n half-edges.*

Proof. This proof will follow a combinatorial approach along with the construction of $CM_n(\mathbf{d})$. Consider the Configuration Model with l_n half-edges and pick a half-edge at random. Then, the selected half-edge can be paired to the other (l_n-1) remaining half-edges. Once an edge has been formed with the first matching, we can continue the same procedure again. We are now left with (l_n-2) half-edges. We pick at random a new half-edge from the remainder. Then this half-edge can be paired to the remaining (l_n-3) half-edges. Once paired, it forms a second edge, and this process can be repeated until all half-edges have been paired. Overall, we have $(l_n-1)\times(l_n-3)\times\ldots\times 3\times 1 = (l_n-1)!!$ possible matchings. □

When studying random graphs, one might be interested in its probability distribution. Before stating the law of $CM_n(\mathbf{d})$, we remark that the Configuration Model is part of the family of multigraphs, since it is allowed to have self-loops and multiple edges (we will have a discussion about simplicity in the next section). Furthermore, for convenience, we define $CM_n(\mathbf{d}) = (X_{ij})_{i,j\in[n]}$ to rewrite the degrees d_i in terms of X_{ij} that gives the number of connections between vertices i and j. This is a way to make self-loops clearly appear, precisely when we have the term X_{ii}. Then, since each self-loop adds 2 to the vertex degree we have:

$$d_i = X_{ii} + \sum_{j\in[n]} X_{ij} = 2X_{ii} + \sum_{i\neq j} X_{ij}. \tag{4.1}$$

We can now state and prove the law of $CM_n(\mathbf{d})$.

Theorem 4.1.2 (Law of $CM_n(\mathbf{d})$)**.** *Let $G = (X_{ij})_{i,j\in[n]}$ be a multigraph on the vertices $[n]$ which is such that:*

$$d_i = X_{ii} + \sum_{j\in[n]} X_{ij}. \tag{4.2}$$

Then,

$$\mathbb{P}(CM_n(\boldsymbol{d}) = G) = \frac{1}{(l_n-1)!!}\frac{\prod_{i\in[n]} d_i!}{\prod_{i\in[n]} 2^{X_{ii}} \prod_{1\leq i\leq j\leq n} X_{ij}!}. \tag{4.3}$$

Proof. We rely on the proof given in [6]. By Theorem 4.1.1, the number of matchings is $(l_n-1)!!$. Then, since each matching has the same probability:

$$\mathbb{P}(CM_n(\mathbf{d}) = G) = \frac{1}{(l_n-1)!!}\alpha(G), \tag{4.4}$$

where $\alpha(G)$ is the number of configurations that give rise to the multigraph G.

Now, if we permute the half-edges incident to a vertex, then the resulting multigraph remains unchanged whilst the configuration is different. There are $\prod_{i\in[n]} d_i!$ ways to permute the half-edges incident to all vertices. However, the permutations of half-edges that contribute to self-loops and multiple edges give rise to the same configurations. Hence, we have to rebalance the multiple countings of self-loops and multiple edges in the number of permutations of half-edges. Now, the number of permutations of half-edges that contribute to self-loops is $\prod_{i\in[n]} 2^{X_{ii}}$, and similarly the number for the multiple edges is $\prod_{1\leq i\leq j\leq n} X_{ij}!$.

Therefore, without multiple countings:

$$\alpha(G) = \frac{\prod_{i\in[n]} d_i!}{\prod_{i\in[n]} 2^{X_{ii}} \prod_{1\leq i\leq j\leq n} X_{ij}!}. \tag{4.5}$$

□

4.2 Regularity Conditions for Vertex Degrees

This section aims to set up some notation and conditions for the theory that will follow.
Now, although **d** is deterministic, we define D to be the degree distribution of the graph and we set n_k to be the number of vertices with degree k. Then the degree distribution in question is defined by the sequence $(p_k)_{k=0}^{\infty}$ given by $p_k = \frac{n_k}{n}$. As we are now considering D as a random variable we can calculate its first and second moment. We then note that:

$$l_n = \sum_{i\in[n]} d_i = \sum_{k=0}^{\infty} kn_k = n\sum_{k=0}^{\infty} kp_k = n\mathbb{E}(D) \Longrightarrow \mathbb{E}(D) = \frac{l_n}{n}, \tag{4.6}$$

along with,

$$\mathbb{E}(D^2) = \sum_{k=0}^{\infty} k^2 p_k = \sum_{i\in[n]} \frac{d_i^2}{n}. \tag{4.7}$$

Furthermore, we define by $D_n = d_v$ the degree of a uniformly chosen vertex v in $[n]$. Then, the random variable D_n has distribution function F_n given by:

$$F_n(x) = \frac{1}{n}\sum_{j\in[n]} \mathbb{1}_{\{d_j\leq x\}}. \tag{4.8}$$

Finally, for the purpose of this chapter, we consider $\mathbb{E}(D)$ and $\mathbb{E}(D^2)$ to be well-defined and finite quantities, as other cases would require more care in the analysis.
We can now give some conditions for vertex degrees [8, pp.23].

Condition 4.2.1 (Weak convergence of vertex degrees)**.** *There exists a distribution function F such that:*

$$D_n \xrightarrow{d} D, \tag{4.9}$$

where D_n and D have distribution functions F_n and F, respectively. We also assume that $F(0) = 0 \Longrightarrow \mathbb{P}(D \geq 1) = 1$.

Condition 4.2.2 (Convergence of average vertex degrees)**.**

$$\lim_{n\to+\infty} \mathbb{E}(D_n) = \mathbb{E}(D) < \infty, \tag{4.10}$$

where D_n and D have distribution functions F_n and F, respectively.

Condition 4.2.3 (Convergence of second moment vertex degrees)**.**

$$\lim_{n\to+\infty} \mathbb{E}(D_n^2) = \mathbb{E}(D^2) < \infty, \tag{4.11}$$

where D_n and D have distribution functions F_n and F, respectively.

4.3 Simplicity of $CM_n(\mathbf{d})$

We recall that the aim of the chapter is to prove that the CM locally converges to a unimodular branching process tree. Since we presented the CM as a multigraph, one might worry that a high number of self-loops and multiples edges would break the condition of local convergence to a tree, which is a structure without any self-loops or multiple edges. We say that a graph is simple if it does not have any self-loops nor multiple edges. Hence, we will engage in a discussion of whether one should be concerned about the non-simplicity of $CM_n(\mathbf{d})$.

4.3.1 Expected Number of Self-loops and Multiple Edges

In this subsection, we will show that when the random graph gets large (when $n \longrightarrow \infty$), the proportion of self-loops and multiple edges becomes small. In other words, we will study the rarity of self-loops and multiple edges in the CM, as the number of vertices approaches infinity.

Let's first set up some notation. Let S_n be the number of self-loops and M_n be the number of multiple edges. Then, using the same notation as in [11];

$$S_n = \sum_{i\in[n]} X_{ii}, \tag{4.12}$$

and,

$$M_n = \sum_{1\le i<j\le n} (X_{ij}-1)_+ = \sum_{1\le i<j\le n} \max\{0,(X_{ij}-1)\}. \tag{4.13}$$

We explain equation (4.13) with a case by case analysis. If $X_{ij}=0$, then i and j are not connected, hence we do not have any multiple edges. If $X_{ij}=1$, then i and j are connected by a single edge, which again implies that we do not have any multiple edges. Finally, if $X_{ij}=\beta\geq 2$, then i and j are connected by β edges, thus it has $\beta-1$ multiple edges. Overall, if $X_{ij}=\{0,1\}$ then $M_n=0$ and if $X_{ij}=\beta\geq 2$ then $M_n=\beta-1$. Note that we can also write $M_n=\frac{1}{2}\sum_{1\le i\neq j\le n}(X_{ij}-1)\mathbb{1}_{\{X_{ij}\geq 2\}}$.
We are now ready to state and prove the following result:

Theorem 4.3.1. *[6, pp.227] Consider S_n and M_n to be the number of self-loops and multiple edges respectively. We define the following quantity:*

$$\nu = \frac{\mathbb{E}[D(D-1)]}{\mathbb{E}[D]} = \frac{\mathbb{E}(D^2)-\mathbb{E}(D)}{\mathbb{E}(D)}. \tag{4.14}$$

Then in $CM_n(\boldsymbol{d})$:

$$\mathbb{E}(S_n) = \frac{1}{2}\sum_{i\in[n]} \frac{d_i(d_i-1)}{l_n-1} \approx \frac{\nu}{2}, \tag{4.15}$$

and:

$$\mathbb{E}(M_n) \le \frac{1}{4}\sum_{1\le i\neq j\le n} \frac{d_i(d_i-1)d_j(d_j-1)}{(l_n-1)(l_n-3)} \approx \frac{\nu^2}{4}. \tag{4.16}$$

Proof. The general steps for the proof are from [6]. We start with the expected number of self-loops. Let $\gamma_{st,i}$ be the indicator that half-edge s is connected to half-edge t where s and t are incident to vertex i. In this proof we are allowed to label the half-edges arbitrarily, where we consider $s,t \in [d_i]$, and label by (i,s) the half-edge s incident to vertex i. Then:

$$S_n = \sum_{i\in[n]} X_{ii} = \sum_{i\in[n]} \sum_{1\le s<t\le d_i} \gamma_{st,i}. \tag{4.17}$$

Therefore we have:

$$\mathbb{E}(S_n) = \sum_{i\in[n]} \sum_{1\le s<t\le d_i} \mathbb{P}(\gamma_{st,i}) = \sum_{i\in[n]} \frac{d_i(d_i-1)}{2}\mathbb{P}(\gamma_{12,i}) = \frac{1}{2}\sum_{i\in[n]} \frac{d_i(d_i-1)}{l_n-1}. \tag{4.18}$$

Where we used the fact that the probability to create a self-loop does not depend on s and t, and also the fact that the probability of $(i,1)$ and $(i,2)$ being connected is $\frac{1}{l_n-1}$. Now, to complete the statement, we note:

$$\mathbb{E}(S_n) = \frac{1}{2}\sum_{i\in[n]} \frac{d_i(d_i-1)}{l_n-1} \approx \frac{1}{2}\sum_{i\in[n]} \frac{d_i(d_i-1)}{l_n} = \frac{1}{2}\sum_{i\in[n]} \frac{\left(\frac{d_i^2}{n}-\frac{d_i}{n}\right)}{\left(\frac{l_n}{n}\right)} = \frac{1}{2}\frac{\mathbb{E}(D^2)-\mathbb{E}(D)}{\mathbb{E}(D)} = \frac{\nu}{2}. \tag{4.19}$$

With a similar argument, we continue with the expected number of multiple edges. Let $\gamma_{s_1t_1,s_2t_2,ij}$ be the indicator that half-edge s_1 is connected to half-edge t_1 and half-edge s_2 is connected to half-edge t_2, where s_1 and s_2 are incident to vertex i, as well t_1 and t_2 are incident to vertex j. Then:

$$M_n = \frac{1}{2}\sum_{1\le i\ne j\le n} (X_{ij}-1)\mathbb{1}_{\{X_{ij}\ge 2\}} \le \frac{1}{2}\sum_{1\le i\ne j\le n} \binom{X_{ij}}{2} = \frac{1}{2}\sum_{j\in[n],i\ne j} \sum_{1\le s_1<s_2\le d_i} \sum_{1\le t_1\ne t_2\le d_j} \gamma_{s_1t_1,s_2t_2,ij}. \tag{4.20}$$

Therefore we have:

$$\begin{aligned}\mathbb{E}(M_n) \le \frac{1}{2}\sum_{1\le i\ne j\le n} \sum_{1\le s_1<s_2\le d_i} \sum_{1\le t_1\ne t_2\le d_j} \mathbb{P}(\gamma_{s_1t_1,s_2t_2,ij}) &= \frac{1}{2}\sum_{1\le i\ne j\le n} \frac{d_i(d_i-1)d_j(d_j-1)}{2}\mathbb{P}(\gamma_{11,22,ij}) \\ &= \frac{1}{4}\sum_{1\le i\ne j\le n} \frac{d_i(d_i-1)d_j(d_j-1)}{(l_n-1)(l_n-3)}.\end{aligned} \tag{4.21}$$

Where we have used the fact that the probability to have multiple edges does not depend on s_i and t_i, and also the fact that the probability of $\{(i,1),(i,2)\}$ and $\{(j,1),(j,2)\}$ being connected is $\frac{1}{(l_n-1)(l_n-3)}$. Now, to complete the statement, we note:

$$\begin{aligned}\mathbb{E}(M_n) &\le \frac{1}{4}\sum_{1\le i\ne j\le n} \frac{d_i(d_i-1)d_j(d_j-1)}{(l_n-1)(l_n-3)} \\ &\approx \frac{1}{4}\sum_{1\le i\ne j\le n} \frac{d_i(d_i-1)d_j(d_j-1)}{l_n^2} = \frac{1}{4}\sum_{1\le i\ne j\le n} \frac{\left(\frac{d_i^2}{n}-\frac{d_i}{n}\right)}{\left(\frac{l_n}{n}\right)}\frac{\left(\frac{d_j^2}{n}-\frac{d_j}{n}\right)}{\left(\frac{l_n}{n}\right)} = \frac{\nu^2}{4}.\end{aligned} \tag{4.22}$$

□

Corollary 4.3.2. *In $CM_n(\boldsymbol{d})$, provided ν is fixed, the proportion of self-loops and multiple edges converges to zero.*

Proof. The general steps for the proof are from [12]. Provided ν is fixed and considering Theorem 4.3.1, we have that $\mathbb{E}(S_n) \approx \frac{\nu}{2}$ and $\mathbb{E}(M_n) \approx \frac{\nu^2}{4}$. Now, it is known that we can express an expectation value in terms of the tail probability so long as we have a non-negative random variable. Hence for S_n we have:

$$\mathbb{E}(S_n) = \sum_{k=1}^{\infty} k\mathbb{P}(S_n = k) = \sum_{k=1}^{\infty}\sum_{\zeta=1}^{k} \mathbb{P}(S_n = k) = \sum_{\zeta=1}^{\infty}\sum_{k=\zeta}^{\infty} \mathbb{P}(S_n = k) = \sum_{\zeta=1}^{\infty} \mathbb{P}(S_n \geq \zeta). \tag{4.23}$$

Now, as a result of (4.23) we state the following lower bound on the expectation of a non-negative random variable X:

$$\mathbb{E}(X) \geq \sum_{n\geq 1} \mathbb{P}(X > n). \tag{4.24}$$

Hence, we can now consider the above X to be $X := \frac{S_n}{\varepsilon}$. Then, for all $\varepsilon > 0$, we have:

$$\frac{\mathbb{E}(S_n)}{\varepsilon} \geq \sum_{n\geq 1} \mathbb{P}\Big(\frac{S_n}{\varepsilon} > n\Big) \iff \frac{\mathbb{E}(S_n)}{\varepsilon} \geq \sum_{n\geq 1} \mathbb{P}\Big(\frac{S_n}{n} > \varepsilon\Big). \tag{4.25}$$

Now, since we assumed that ν is fixed, with (4.25) we see that $\frac{\mathbb{E}(S_n)}{\varepsilon} < \infty \Longrightarrow \sum_{n=1}^{\infty} \mathbb{P}(\frac{S_n}{n} > \varepsilon) < \infty$. And by the Borel-Cantelli Lemma, $\mathbb{P}(\frac{S_n}{n} > \varepsilon \text{ i.o.}) = 0$.
Finally, recall that a criterion for almost sure (a.s.) convergence is:

$$\sum_{n=1}^{\infty} \mathbb{P}\big(|X_n - X| > \varepsilon\big) < \infty \Longrightarrow X_n \xrightarrow{a.s.} X. \tag{4.26}$$

Then as a result of Borel-Cantelli:

$$\sum_{n=1}^{\infty} \mathbb{P}\Big(\Big|\frac{S_n}{n}\Big| > \varepsilon\Big) < \infty \Longrightarrow \frac{S_n}{n} \xrightarrow{a.s.} 0. \tag{4.27}$$

This proves that the proportion of self-loops in $CM_n(\mathbf{d})$ converges to zero as $n \to +\infty$. The proof for the proportion of multiple edges is the same, just replacing S_n by M_n. The result will be: $\frac{M_n}{n} \xrightarrow{a.s.} 0$. □

4.3.2 Probability of Simplicity

Although the study on the expected number of self-loops and multiple edges is an interesting analysis to give an idea about what we could expect in a specific configuration. It would also be another interesting fact to give a concrete result for the probability of simplicity in the CM.
We aim to present and sketch the proof of the following theorem:

Theorem 4.3.3 (Probability of simplicity of $CM_n(\mathbf{d})$)**.** *Assume conditions 4.2.1, 4.2.2 and 4.2.3 hold. Then, with* $\nu = \frac{\mathbb{E}[D(D-1)]}{\mathbb{E}[D]}$,

$$\lim_{n\to+\infty} \mathbb{P}(CM_n(\boldsymbol{d}) \text{ is simple}) = e^{-\frac{\nu}{2}-\frac{\nu^2}{4}}. \tag{4.28}$$

As a matter of space and relevance, we will only give a sketch of the proof for Theorem 4.3.3, where the details can be found in [6].

Sketch of Proof. For the purpose of this proof, set $(X)_r := X(X-1)...(X-r+1)$ and define the r-th factorial moment of X to be $\mathbb{E}[(X)_r]$. Furthermore we admit without proof that a vector of integer-valued random variables $\left((X_n^1, X_n^2, ..., X_n^d)\right)_{n\geq 1}$ converges in distribution to a vector of independent Poisson random variables with parameters $\lambda_1, ..., \lambda_d$ when, for all $r_1, ..., r_d \in \mathbb{N}$,

$$\lim_{n\to+\infty} \mathbb{E}\left[(X_n^1)_{r_1}...(X_n^d)_{r_d}\right] = \lambda_1^{r_1}...\lambda_d^{r_d}. \tag{4.29}$$

Now, recall the formalism of S_n and M_n given in equation (4.17) and (4.20). Hence, the aim is to show that the vector (S_n, M_n) converges in distribution to (S, M), where S and M are two independent Poisson random variables with parameters $\frac{\nu}{2}$ and $\frac{\nu^2}{4}$ respectively. Therefore, by equation (4.29) we aim to prove that:

$$\lim_{n\to+\infty} \mathbb{E}\left[(S_n)_u (M_n)_w\right] = \left(\frac{\nu}{2}\right)^u \left(\frac{\nu^2}{4}\right)^w. \tag{4.30}$$

For this, we will proceed by a sandwiching argument of (4.30). Now for convenience, we define the following sets of vertices for the creation of self-loops and multiple edges:

$$\Lambda_1 = \{(st, i) : i \in [n], 1 \leq s < t \leq d_i\}, \tag{4.31}$$

and,

$$\Lambda_2 = \{(s_1 t_1, s_2 t_2, ij) : 1 \leq i < j \leq n, 1 \leq s_1 < s_2 \leq d_i, 1 \leq t_1 \neq t_2 \leq d_j\}. \tag{4.32}$$

Since the vertices with degree at least 2 are the only vertices that contribute to Λ_1 and Λ_2, we can assume that the cardinalities $|\Lambda_1|$ and $|\Lambda_2|$ grow to infinity as $n \to +\infty$. We can further assume that under Condition 4.2.1, 4.2.2 and 4.2.3, $\max_{i\in[n]} d_i = o(\sqrt{n})$ and $l_n \geq n$. Then, using similar combinatorial arguments as for the proof of Theorem 4.3.1 and finding an upper bound for $\mathbb{E}[(S_n)_u (M_n)_w]$ in the case $\nu = 0$, we have:

$$\limsup_{n\to+\infty} \mathbb{E}[(S_n)_u (M_n)_w] \leq \left(\lim_{n\to+\infty} \frac{|\Lambda_1|}{l_n}\right)^u \left(\lim_{n\to+\infty} \frac{|\Lambda_2|}{l_n^2}\right)^w. \tag{4.33}$$

Under the given conditions, where u and w are fixed, we can calculate the limits on the right hand side of equation (4.33), and we find:

$$\lim_{n\to+\infty} \frac{|\Lambda_1|}{l_n} = \frac{\nu}{2} \quad \text{and} \quad \lim_{n\to+\infty} \frac{|\Lambda_2|}{l_n^2} = \left(\frac{\nu}{2}\right)^2. \tag{4.34}$$

Finally, substituting these results in equation (4.33) we obtain the required upper bound:

$$\limsup_{n\to+\infty}\mathbb{E}[(S_n)_u(M_n)_w] \leq \left(\frac{\nu}{2}\right)^u \left(\frac{\nu^2}{4}\right)^w \Longrightarrow \lim_{n\to+\infty}\mathbb{E}[(S_n)_u(M_n)_w] \leq \left(\frac{\nu}{2}\right)^u \left(\frac{\nu^2}{4}\right)^w. \tag{4.35}$$

Using similar arguments in the case $\nu > 0$, one can show that:

$$\lim_{n\to+\infty}\mathbb{E}[(S_n)_u(M_n)_w] \geq \left(\frac{\nu}{2}\right)^u \left(\frac{\nu^2}{4}\right)^w, \tag{4.36}$$

and overall, for any $\nu \geq 0$, we have shown that:

$$\lim_{n\to+\infty}\mathbb{E}[(S_n)_u(M_n)_w] = \left(\frac{\nu}{2}\right)^u \left(\frac{\nu^2}{4}\right)^w \Longrightarrow (S_n, M_n) \xrightarrow{d} (S, M). \tag{4.37}$$

Then, since S and M are independent Poisson random variables, they have the following probability mass functions:

$$\mathbb{P}(S=\alpha) = \frac{\left(\frac{\nu}{2}\right)^\alpha e^{-\frac{\nu}{2}}}{\alpha!} \quad \text{and} \quad \mathbb{P}(M=\beta) = \frac{\left(\frac{\nu^2}{4}\right)^\beta e^{-\frac{\nu^2}{4}}}{\beta!}. \tag{4.38}$$

Hence, by independence:

$$\begin{aligned}\lim_{n\to+\infty}\mathbb{P}(CM_n(\mathbf{d}) \text{ is simple}) = \lim_{n\to+\infty}\mathbb{P}(S_n = M_n = 0) &= \left(\lim_{n\to+\infty}\mathbb{P}(S_n=0)\right)\left(\lim_{n\to+\infty}\mathbb{P}(M_n=0)\right) \\ &= \mathbb{P}(S=0)\mathbb{P}(M=0) \\ &= e^{-\frac{\nu}{2}-\frac{\nu^2}{4}}.\end{aligned} \tag{4.39}$$

□

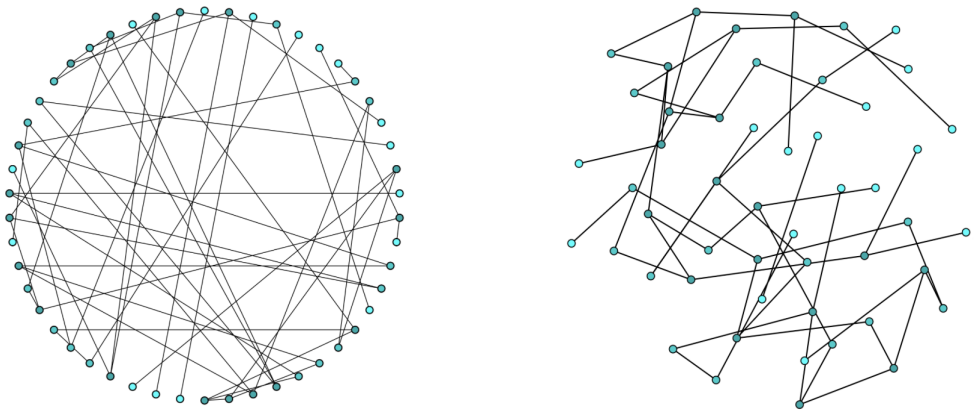


Figure 4.3: Representation of $CM_n(\mathbf{d})$ in the case $n = 50$ and $d_i = \{1,2,3\}$. In this case, we observe that the resulting graph does not have any self-loops or multiple edges. [This figure has been generated using The Network Pages(2024)]

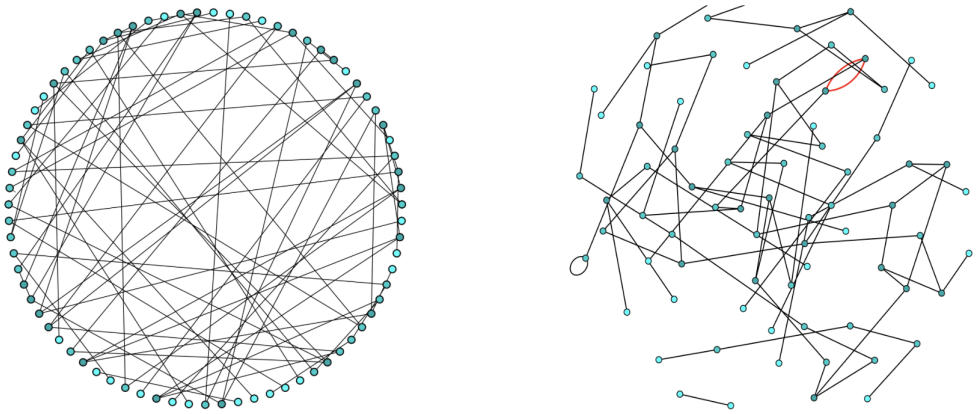


Figure 4.4: Representation of $CM_n(\mathbf{d})$ in the case $n = 75$ and $d_i = \{1,2,3\}$. In this case, we observe that the resulting graph has one self-loop and one multiple edge. [This figure has been generated using The Network Pages(2024)]

Remark 4.3.4. *A common variant of the Configuration Model is the Erased Configuration Model, where self-loops are removed and multiple edges are merged. It can be shown that, as $n \to +\infty$, the degree distribution of the Erased Configuration Model converges to that of the Configuration Model. This is a simple method to generate simple configurations.*

4.4 The Local Limit of $CM_n(\mathbf{d})$

In this section, we will prove the main result of this chapter which can be stated as the following theorem, adapted from [8, pp.136]:

Theorem 4.4.1 (Locally tree-like nature of the CM)**.** *Assume conditions 4.2.1 and 4.2.2 hold. Then, in the case where vertices have degrees 1 or 3, $CM_n(\boldsymbol{d})$ converges locally in probability to the unimodular branching process tree with root offspring distribution $(p_k)_{k\geq 0}$ given by $p_k = \mathbb{P}(D = k)$.*

For the duration of this chapter we will refer to $\mathbf{d}^*$ for the prescribed degree sequence with $d_i = \{1,3\}$.

4.4.1 Setup and Simplicity of $CM_n(\mathbf{d}^*)$

For this section, we will define and analyse the CM in its general form with the constrained degree sequence $\mathbf{d}^*$. For this, we consider $\alpha \in (0,1)$ such that we have αn vertices with degree 1 and $(1-\alpha)n$ vertices with degree 3. Furthermore we require the sum of the degrees l_n to be even, and with this setup, it implies that n has to be even. Then, if we consider $2m$ half-edges in $CM_n(\mathbf{d}^*)$, we have:

$$l_n = \sum_{i\in[n]} d_i = \alpha n + 3n(1-\alpha) = 3n - 2\alpha n = n(3-2\alpha) = 2m. \tag{4.40}$$

We also note that the root offspring distribution will only have two possibilities, p_1 and p_3. To find these probabilities, we recall the beginning of section 4.2 where we defined $p_k = \frac{n_k}{n}$. Now using the setup for $\mathbf{d}^*$, we have that $p_1 = \frac{n\alpha}{n} = \alpha$, and $p_3 = \frac{n(1-\alpha)}{n} = 1-\alpha$.

Remark 4.4.2. *For convenience, we have simplified notation in the setup so that we do not carry floor functions in the rest of the chapter. However, for completeness, we need to keep in mind that not considering the floor notation might give rise to a rational number of vertices, and we should really be considering* $\lfloor \alpha n \rfloor$ *vertices with degree* 1 *and* $\lfloor (1-\alpha)n \rfloor$ *vertices with degree* 3*. Regardless of this subtlety, we will consider that we have a good enough approximation by defining* $p_1 = \frac{\lfloor \alpha n \rfloor}{n} \approx \alpha$ *and* $p_3 = \frac{\lfloor (1-\alpha)n \rfloor}{n} \approx 1-\alpha$*, and we will continue this section without the floor functions.*

We will now have a brief discussion about the expected number of self-loops and multiple edges, and the simplicity of $CM_n(\mathbf{d}^*)$.
First note:

$$\mathbb{E}(D) = \frac{l_n}{n} = 3 - 2\alpha, \tag{4.41}$$

and,

$$\mathbb{E}(D^2) = \sum_{i \in [n]} \frac{d_i^2}{n} = \frac{\alpha n + 9n(1-\alpha)}{n} = 9 - 8\alpha. \tag{4.42}$$

Then, finding an expression for ν we have:

$$\nu = \frac{\mathbb{E}(D^2) - \mathbb{E}(D)}{\mathbb{E}(D)} = \frac{6 - 6\alpha}{3 - 2\alpha}. \tag{4.43}$$

Now using the approximations in Theorem 4.3.1, the expected number of self-loops is:

$$\mathbb{E}(S_n) \approx \frac{\nu}{2} = \frac{3 - 3\alpha}{3 - 2\alpha}. \tag{4.44}$$

And for the expected number of multiple edges:

$$\mathbb{E}(M_n) \leq \frac{\nu^2}{4} = \frac{1}{4}\left(\frac{6 - 6\alpha}{3 - 2\alpha}\right)^2. \tag{4.45}$$

Then, to make sense of these results, we first have that both $\mathbb{E}(S_n)$ and $\mathbb{E}(M_n)$ are strictly decreasing functions for $\alpha \in (0,1)$. Now, if the random graph tends to have more vertices with degree 1 (i.e. $\alpha \to 1$), then both expected numbers of self-loops and multiple edges converges to 0. On the other hand, if the random graph tends to have more vertices with degree 3 (i.e. $\alpha \to 0$) then both expected numbers of self-loops and multiple edges converges to 1.

Furthermore using Theorem 4.3.3, the probability of simplicity $\mathbb{P}^*$ is given by:

$$\mathbb{P}^* = \lim_{n \to +\infty} \mathbb{P}(CM_n(\mathbf{d}^*) \text{ is simple}) = e^{-\frac{\nu}{2} - \frac{\nu^2}{4}}. \tag{4.46}$$

Substituting equation (4.43) in (4.46), we have that:

$$e^{-2} < \mathbb{P}^* < 1. \tag{4.47}$$

Again, to make sense of this result, we first observe that $\mathbb{P}^*$ is a strictly increasing function on

$\alpha \in (0,1)$. So if the random graph tends to have more vertices with degree 1 (i.e. $\alpha \to 1$) then it is more likely to be simple, with $\mathbb{P}^* \to 1$. On the other hand, if the random graph tends to have more vertices with degree 3 (i.e. $\alpha \to 0$) then it is more likely to be non-simple, with $\mathbb{P}^* \to e^{-2} \approx 0.14$.

The overall key point is that, as $n \to +\infty$, the proportion of self-loops and multiple edges approaches 0. Additionally, even if we find a relatively small number of self-loops and multiple edges, they will actually not have an impact on the local limit to a tree. Without loss of generality, from now on we can condition $CM_n(\mathbf{d}^*)$ on simplicity and disregard any a priori concerns on the non-simplicity of the model.

4.4.2 Convergence of $CM_n(\mathbf{d}^*)$ - An Example up to Radius 2

In this subsection, we will prove the convergence of the *CM* in a specific example with a degree sequence $\mathbf{d}^*$. As a first approach, we will prove Theorem 4.4.1 up to radius 2, where we consider a rooted graph with two degree 3 vertices and four degree 1 vertices.

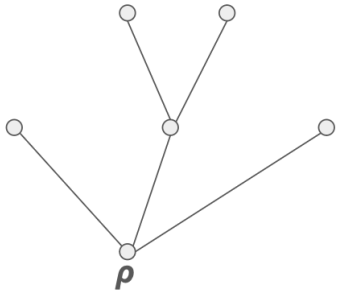


Figure 4.5: Representation of the rooted tree (H, ρ_H) up to radius 2. [This figure has been created using GeoGebra(2024)]

Considering Definition 3.2.2, our aim is to show that:

$$\mathbb{P}\Big(B_2^{CM_n(\mathbf{d}^*)}(\rho_n) \cong (H,\rho_H)\Big|CM_n(\mathbf{d}^*)\Big) \xrightarrow{\mathbb{P}} \mathbb{P}\Big(B_2^{BP_U}(\rho) \cong H\Big), \tag{4.48}$$

where BP_U is to denote the unimodular branching process.
To understand (4.48) we can reformulate the equation in words. To show this limit, we are asked to prove that the probability that ρ_H in the *CM* looks like H up to radius 2 is approximately equal to the probability that the branching process BP_U gives H.
Recall that $p_1 = \alpha$, $p_3 = 1-\alpha$ and $\mathbb{E}(D) = 3-2\alpha$. We note that $n_i d_i$ corresponds to the number of half-edges incident to all the vertices with degree i.
Now let,

$$X := \mathbb{P}\Big(B_2^{CM_n(\mathbf{d}^*)}(\rho_n) \cong (H,\rho_H)\Big|CM_n(\mathbf{d}^*)\Big), \tag{4.49}$$

and,

$$X^* := \mathbb{P}\Big(B_2^{BP_U}(\rho) \cong H\Big). \tag{4.50}$$

Hence, by the definition of convergence in probability, we need to prove that for all $\varepsilon > 0$:

$$\lim_{n\to+\infty} \mathbb{P}(|X - X^*| > \varepsilon) = 0. \tag{4.51}$$

Now looking at Figure 4.5, we can obtain an expression for X using a combinatorial argument with the following steps: first, we pick a root at random in $CM_n(\mathbf{d}^*)$ and suppose that we end up on a vertex with degree 3 (see Remark 4.4.3 for the choice of root). This event occurs with probability p_3. Now looking at the given tree H, we observe that the root needs to connect to two degree 1 vertices and one degree 3 vertex (up to radius 1). If we want to connect the root to a degree 3 vertex, we need to pair one of the half-edges together. Among the original l_n half-edges in the model, there are precisely $3(n_3-1)$ choices of half-edges incident to all the vertices with degree 3 (excluding the root). Then we have a probability of $\frac{3(n_3-1)}{l_n-1}$ to pair the root to a degree 3 vertex. Moving forward in the calculations, we are now searching for a degree 1 vertex. Again, among the original l_n half-edges, n_1 of them are incident to all the vertices with degree 1. Since we have already used two half-edges to connect up the root a first time, the probability that we connect the root to a second vertex that has degree 1 is $\frac{n_1}{l_n-3}$. This process continues until all the vertices have been connected to produce H, remembering that at each step two half-edges are being used to connect two vertices.

Considering the above explanation, we can now formally give the following expression for X:

$$\begin{aligned} X &= p_3 \left(\frac{3(n_3-1)}{l_n-1}\right)\left(\frac{n_1}{l_n-3}\right)\left(\frac{n_1-1}{l_n-5}\right)\left(\frac{n_1-2}{l_n-7}\right)\left(\frac{n_1-3}{l_n-9}\right) \\ &= 3(1-\alpha)\left(\frac{n_3-1}{l_n-1}\right)\left(\frac{n_1}{l_n-3}\right)\left(\frac{n_1-1}{l_n-5}\right)\left(\frac{n_1-2}{l_n-7}\right)\left(\frac{n_1-3}{l_n-9}\right), \end{aligned} \tag{4.52}$$

and if we take n large enough, such that $l_n \to +\infty$, we have the following approximation:

$$\begin{aligned} X \approx 3(1-\alpha)\left(\frac{n_3}{l_n}\right)\left(\frac{n_1}{l_n}\right)^4 = 3(1-\alpha)\left(\frac{np_3}{n\mathbb{E}(D)}\right)\left(\frac{np_1}{n\mathbb{E}(D)}\right)^4 &= \frac{3(1-\alpha)^2\alpha^4}{\mathbb{E}(D)^5} \\ &= \frac{3(1-\alpha)^2\alpha^4}{(3-2\alpha)^5}. \end{aligned} \tag{4.53}$$

Our next task is to find an expression for X^* based on the construction of the unimodular branching process given in Section 2.3.2 . We recall that in a unimodular setup, the offspring distribution of the root is given by p_k while all other vertices have root offspring distribution p_k^* described by the size-biased version of D given by: $p_k^* = \mathbb{P}(D^* - 1 = k) = \frac{(k+1)p_{k+1}}{\mathbb{E}(D)}$, where we see that p_k^* will only take the values $k = 0$ and $k = 2$.

To find an expression for X^*: we fix the root to be a vertex of degree 3, with probability p_3, and we complete the calculations with the remaining vertices in the tree. In our case, we require four vertices with degree 1 and an extra vertex with degree 3, where the offspring distributions are given by p_0^* and p_2^* respectively. Hence, we find the following:

$$X^* = p_3 p_2^* (p_0^*)^4 = (1-\alpha)\left(\frac{3p_3}{\mathbb{E}(D)}\right)\left(\frac{p_1}{\mathbb{E}(D)}\right)^4 = \frac{3(1-\alpha)^2\alpha^4}{\mathbb{E}(D)^5} = \frac{3(1-\alpha)^2\alpha^4}{(3-2\alpha)^5}. \tag{4.54}$$

Overall, in the limit we have that $|X - X^*| \to 0$ which implies $X \xrightarrow{\mathbb{P}} X^*$.

Remark 4.4.3 (Choice of root)**.** *In this section we have randomly chosen the root to be a vertex of degree 3. However, adapting the expression for X and X^* in the case where the root is a vertex of degree 1 and using similar combinatorial arguments as before, we obtain:*

$$X = p_1 \left(\frac{3n_3}{l_n - 1}\right)\left(\frac{3(n_3-1)}{l_n-3}\right)\left(\frac{n_1-2}{l_n-5}\right)\left(\frac{n_1-3}{l_n-7}\right)\left(\frac{n_1-4}{l_n-9}\right) \approx \frac{9(1-\alpha)^2\alpha^4}{(3-2\alpha)^5}, \tag{4.55}$$

$$X^* = p_1(p_2^*)^2(p_0^*)^3 = \frac{9(1-\alpha)^2\alpha^4}{(3-2\alpha)^5}, \tag{4.56}$$

where again the approximation in (4.55) comes from taking $l_n \to +\infty$. In this instance, we remark that choosing the root at random does not affect the local limit and we conclude once more that $|X - X^*| \to 0$ which implies $X \xrightarrow{\mathbb{P}} X^*$.

4.4.3 Convergence of $CM_n(\mathbf{d}^*)$ - A General Proof up to Radius r

We now aim to prove Theorem 4.4.1 in its full generality up to an arbitrary radius r. We then consider a rooted tree H with ω_3 vertices with degree 3 and ω_1 vertices with degree 1, where the radius around the root is at most r.

Considering again Definition 3.2.2, our aim is to show that:

$$\mathbb{P}\Big(B_r^{CM_n(\mathbf{d}^*)}(\rho_n) \cong (H,\rho_H)\Big|CM_n(\mathbf{d}^*)\Big) \xrightarrow{\mathbb{P}} \mathbb{P}\Big(B_r^{BP_U}(\rho) \cong H\Big), \tag{4.57}$$

where BP_U is still to denote the unimodular branching process.

To show this limit, we are asked to prove that the probability that ρ_H in the CM looks like H up to radius r is approximately equal to the probability that the branching process BP_U gives H.

We also define:

$$Y := \mathbb{P}\Big(B_r^{CM_n(\mathbf{d}^*)}(\rho_n) \cong (H,\rho_H)\Big|CM_n(\mathbf{d}^*)\Big), \tag{4.58}$$

and,

$$Y^* := \mathbb{P}\Big(B_r^{BP_U}(\rho) \cong H\Big). \tag{4.59}$$

Similarly to the example above, we aim to show that $|Y - Y^*| \to 0$.

We now explain how to calculate Y and Y^*. Recall that we are considering $CM_n(\mathbf{d}^*)$ where we have n_3 vertices with degree 3 and n_1 vertices with degree 1. We note that $\omega_1 \le n_1$ and $\omega_3 \le n_3$, and for convenience we set $\Phi = \omega_1 + \omega_3$.

Starting with the calculations for X, we explain its construction using a combinatorial argument. We first arbitrarily fix the root to be a vertex with degree 3, which occurs with probability p_3. Now our next task is to connect the root to the remaining $\Phi - 1$ vertices from the tree H. For this, we consider the two scenarios of connecting to a degree 3 vertex, and then connecting to a degree 1 vertex. Recall that every time we connect two vertices together, we have a depletion of 2 half-edges from l_n.

1) In the first scenario, among the original l_n half-edges in the model, there are precisely $3(n_3-1)$ choices of half-edges incident to all the vertices with degree 3 (excluding the root). Then we have a probability of $\frac{3(n_3-1)}{l_n-1}$ to pair the root to a degree 3 vertex. Furthermore, if we want to connect to a second degree 3 vertex, we have now $3(n_3-2)$ choices of half-edges incident to the remaining n_3-2 degree 3 vertices. This implies that we have a probability of $\frac{3(n_3-2)}{l_n-3}$ of this occurring. In generality (excluding the root), the probability to connect ω_3-1 vertices to the root is:

$$\prod_{i=1}^{\omega_3-1}\prod_{k=1}^{\omega_3-1}\frac{3(n_3-k)}{l_n-(2i-1)}. \tag{4.60}$$

2) In the second scenario, we are left with $l_n-(2\omega_3-1)$ half-edges in the model. There are precisely n_1 choices of half-edges incident to all the vertices with degree 1. Then we have a probability of $\frac{n_1}{l_n-(2\omega_3-1)}$ to connect to a degree 1 vertex. Furthermore, if we want to connect to a second degree 1 vertex, we have now n_1-1 choices of half-edges incident to the remaining n_1-1 degree 1 vertices. This implies that we have a probability of $\frac{n_1-1}{l_n-(2\omega_3+1)}$ of this happening. In general, the probability to connect ω_1 vertices is:

$$\prod_{i=\omega_3}^{\Phi-1}\prod_{j=0}^{\omega_1-1}\frac{n_1-j}{l_n-(2i-1)}. \tag{4.61}$$

Based on the construction above, we can now formally give the following expression for Y:

$$\begin{aligned} Y &= p_3\prod_{i=1}^{\Phi-1}\prod_{j=0}^{\omega_1-1}\prod_{k=1}^{\omega_3-1}\left[\frac{3(n_3-k)(n_1-j)}{l_n-(2i-1)}\right] \\ &= p_3\prod_{i=1}^{\Phi-1}\left[\frac{1}{l_n-(2i-1)}\right]\prod_{j=0}^{\omega_1-1}(n_1-j)\prod_{k=1}^{\omega_3-1}3(n_3-k). \end{aligned} \tag{4.62}$$

Moreover if we take n large enough, such that $l_n\to+\infty$, we have the following approximation:

$$\begin{aligned} Y\approx p_3\prod_{i=1}^{\Phi-1}\left[\frac{1}{l_n}\right]\prod_{j=0}^{\omega_1-1}(n_1)\prod_{k=1}^{\omega_3-1}(3n_3) &= p_3\frac{(3n_3)^{\omega_3-1}n_1^{\omega_1}}{l_n^{\Phi-1}} \\ &= 3^{\omega_3-1}(1-\alpha)\left(\frac{n_3}{l_n}\right)^{\omega_3-1}\left(\frac{n_1}{l_n}\right)^{\omega_1} \\ &= 3^{\omega_3-1}(1-\alpha)\left(\frac{np_3}{n\mathbb{E}(D)}\right)^{\omega_3-1}\left(\frac{np_1}{n\mathbb{E}(D)}\right)^{\omega_1} \\ &= \frac{3^{\omega_3-1}(1-\alpha)^{\omega_3}\alpha^{\omega_1}}{\mathbb{E}(D)^{\Phi-1}} \\ &= \frac{3^{\omega_3-1}(1-\alpha)^{\omega_3}\alpha^{\omega_1}}{(3-2\alpha)^{\Phi-1}}. \end{aligned} \tag{4.63}$$

Next, we need to find an expression for Y^*. For this, we again rely on the construction of a unimodular branching process given in subsection 2.3.2. We first fix the root of BP_U to be a vertex of degree 3, where this has probability p_3 of happening. Next, by definition, the probability of the offsprings in later generations is given by p_0^* for vertices with degree 1, and by p_2^* for vertices with degree 3. Then,

excluding the root, we need to connect ω_1 vertices with degree 1 and $\omega_3 - 1$ vertices with degree 3. Hence we find the following:

$$\begin{aligned} Y^* = p_3 \prod_{i=1}^{\omega_3 - 1} (p_2^*) \prod_{j=1}^{\omega_1} (p_0^*) = p_3 (p_2^*)^{\omega_3 - 1} (p_0^*)^{\omega_1} &= (1-\alpha) \left(\frac{3p_3}{\mathbb{E}(D)} \right)^{\omega_3 - 1} \left(\frac{p_1}{\mathbb{E}(D)} \right)^{\omega_1} \\ &= \frac{3^{\omega_3 - 1} (1-\alpha)^{\omega_3} \alpha^{\omega_1}}{\mathbb{E}(D)^{\Phi - 1}} \\ &= \frac{3^{\omega_3 - 1} (1-\alpha)^{\omega_3} \alpha^{\omega_1}}{(3-2\alpha)^{\Phi - 1}}. \end{aligned} \tag{4.64}$$

Overall we have that $|Y - Y^*| \to 0$ which implies $Y \xrightarrow{\mathbb{P}} Y^*$, and proves Theorem 4.4.1 in its full generality.

Remark 4.4.4 (Choice of root)**.** *We again make a similar remark as in the previous example on the choice of root. In this section, we considered the root to be a vertex of degree 3. However, if we adapt the expressions for Y and Y^* in the case where the root has degree 1, we find the following:*

$$Y = p_1 \prod_{i=1}^{\Phi - 1} \prod_{j=1}^{\omega_1 - 1} \prod_{k=0}^{\omega_3 - 1} \left[\frac{3(n_3 - k)(n_1 - j)}{l_n - (2i-1)} \right] \approx \frac{3^{\omega_3} (1-\alpha)^{\omega_3} \alpha^{\omega_1}}{(3-2\alpha)^{\Phi - 1}}, \tag{4.65}$$

$$Y^* = p_1 \prod_{i=1}^{\omega_3} (p_2^*) \prod_{j=1}^{\omega_1 - 1} (p_0^*) = p_1 (p_2^*)^{\omega_3} (p_0^*)^{\omega_1 - 1} = \frac{3^{\omega_3} (1-\alpha)^{\omega_3} \alpha^{\omega_1}}{(3-2\alpha)^{\Phi - 1}}, \tag{4.66}$$

where again the approximation in (4.65) comes from taking $l_n \to +\infty$. In this instance, we again remark that choosing the root at random does not affect the local limit and we conclude once more that $|Y - Y^*| \to 0$ which implies $Y \xrightarrow{\mathbb{P}} Y^*$.

Chapter 5

The Giant Component

We will end this thesis on the study of the greatest component by precisely looking at how the local limit can help understand properties of the giant component. We will first investigate the giant as an almost local property of the random graph before shifting our interest to an application involving the largest component in an Erdős-Rényi Random Graph $ER_n(\frac{\lambda}{n})$.

Considering two vertices u and v in $[n]$, we write $u \longleftrightarrow v$ when these two vertices are connected by a path of vertices.
For $v \in [n]$, we define the *connected component containing v* by:

$$\mathbf{C}(v) = \{u \in [n] : v \longleftrightarrow u\}. \tag{5.1}$$

Furthermore, we define the *greatest connected component* $\mathbf{C}_{max}$ to be the largest component for which the cardinality of $\mathbf{C}$ is maximal:

$$|\mathbf{C}_{max}| = \max_{v \in [n]} |\mathbf{C}(v)|. \tag{5.2}$$

5.1 The Giant is Almost Local

Recall that χ is the survival probability of a local limit. Further recall that $\mathbf{C}(v)$ is the connected component around a vertex $v \in [n]$ and that $|\mathbf{C}(v)|$ is the size of this component. Then we may define $\chi = \lim_{k \to +\infty} \chi_{\geq k}$ where $\chi_{\geq k} = \mathbb{P}(|\mathbf{C}(v)| \geq k)$, for all $k \geq 1$.
We aim to prove the following theorem for a general random graph G_n, whose size we assume to be $|V(G_n)| = n$ without loss of generality.
The general results and proofs in this section are from [13], however we will delve deeper in the arguments and carefully explain each step in the proofs.

Theorem 5.1.1. *(The giant is almost local) Let $G_n = ([n], E(G_n))$ denote a random graph of size n. Assume that G_n converges locally in probability to (G, ρ), i.e. following Criterion 3.3.2:*

$$\frac{1}{n} \sum_{v \in [n]} \mathbb{1}_{\{B_r^{G_n}(v) \cong H\}} \xrightarrow{\mathbb{P}} \mathbb{P}(B_r^G(\rho) \cong H) \tag{5.3}$$

Assume further that,

$$\lim_{k\to+\infty} \limsup_{n\to+\infty} \frac{1}{n^2}\mathbb{E}\big[|\{(x,y)\in[n]\times[n] : |\mathbf{C}(x)|, |\mathbf{C}(y)| \geq k, \mathbf{C}(x)\cap\mathbf{C}(y)=\varnothing\}|\big] = 0 \tag{5.4}$$

Then, denoting $\mathbf{C}_2$ *as the second largest component,*

$$\frac{|\mathbf{C}_{max}|}{n} \xrightarrow{\mathbb{P}} \chi \qquad \text{and} \qquad \frac{|\mathbf{C}_2|}{n} \xrightarrow{\mathbb{P}} 0 \tag{5.5}$$

Remark 5.1.2. *The above condition (5.4) will be most useful to explain the uniqueness of the giant. Indeed, it states that if we choose two arbitrary vertices* $(x,y)\in[n]\times[n]$ *and we grow the size of their connected components* $|\mathbf{C}(x)|$ *and* $|\mathbf{C}(y)|$ *to infinity then the event that these two components are disjoint, i.e.* $\{\mathbf{C}(x)\cap\mathbf{C}(y)=\varnothing\}$*, is highly unlikely as* $n\to+\infty$*. And in fact we will be able to conclude that they must be part of the same connected component.*

To prove Theorem 5.1.1, we will consider the two cases where $\chi = 0$ and $\chi > 0$.

5.1.1 Upper Bound on the Size of the Giant

We first start with the statement and proof of a corollary on the size of the giant component. We will then be able to conclude a first result when $\chi = 0$.

Corollary 5.1.3. *(Upper bound on the size of the giant) Let* $(G_n)_{n\geq1}$ *be a sequence of random graphs with size n. Assume that* G_n *converges locally in probability to* (G,ρ)*. Then for* $\chi = \mathbb{P}(|\mathbf{C}(\rho)|=\infty)$ *and for every* $\varepsilon > 0$ *fixed,*

$$\mathbb{P}(|\mathbf{C}_{max}| \leq n(\chi+\varepsilon)) \xrightarrow{n\to+\infty} 1 \tag{5.6}$$

Proof. We first define $\mathbf{C}_{\geq k}$ and $\mathbf{C}_{<k}$ as:

$$\mathbf{C}_{\geq k} = \sum_{v\in[n]} \mathbb{1}_{\{|\mathbf{C}(v)|\geq k\}} \qquad \text{and,} \qquad \mathbf{C}_{<k} = \sum_{v\in[n]} \mathbb{1}_{\{|\mathbf{C}(v)|<k\}} \tag{5.7}$$

Now we assumed that G_n converges locally in probability to (G,ρ) for all rooted graphs (H,ρ_H) with size k, where we note that there is a finite number of such graphs.
Then, using this local limit assumption,

$$\begin{aligned}\frac{1}{n}\sum_{v\in[n]} \mathbb{1}_{\{|\mathbf{C}(\rho_n)|=k\}} &= \frac{1}{n}\sum_{|V(H)|=k}\sum_{v\in[n]} \mathbb{1}_{\{B_r^{G_n}(\rho_n)\cong(H,\rho_H)\}} \\ &\xrightarrow{\mathbb{P}} \sum_{|V(H)|=k} \mathbb{P}(B_r^G(\rho)\cong H) = \mathbb{P}(|G|=k)\end{aligned} \tag{5.8}$$

Recall that if a sequence of random variables $\{X_i\}_{i=1}^n$ converges in probability to limiting random variables $\{Y_i\}_{i=1}^n$, such that $X_i \xrightarrow{\mathbb{P}} Y_i$, then $\sum_{i=1}^n X_i \xrightarrow{\mathbb{P}} \sum_{i=1}^n Y_i$.
Now, using (5.8), for all $k\geq1$ we have that,

$$\sum_{i<k}\frac{1}{n}\sum_{v\in[n]} \mathbb{1}_{\{|\mathbf{C}(\rho_n)|=i\}} = \frac{1}{n}\sum_{v\in[n]} \mathbb{1}_{\{|\mathbf{C}(\rho_n)|<k\}} = \frac{\mathbf{C}_{<k}}{n}, \tag{5.9}$$

and,

$$\sum_{i<k} \mathbb{P}(|G| = i) = \mathbb{P}(|G| < k). \tag{5.10}$$

Therefore, taking complements, we have the following result: $\frac{\mathbf{C}_{<k}}{n} \xrightarrow{\mathbb{P}} \mathbb{P}(|G| < k) \implies \frac{\mathbf{C}_{\geq k}}{n} \xrightarrow{\mathbb{P}} \mathbb{P}(|G| \geq k)$.

We now note that for all $k \geq 1$, $\{|\mathbf{C}_{max}| \geq k\} = \{\mathbf{C}_{\geq k} \geq k\}$. Furthermore, and by definition, $\lim_{k\to+\infty} \mathbb{P}(|G| \geq k) = \chi$ hence, for all $\varepsilon > 0$ we have the following upper bound: $\mathbb{P}(|G| \geq k) < \chi + \varepsilon$.
Therefore, $\{|\mathbf{C}_{max}| \geq n(\chi+\varepsilon)\} = \{|\mathbf{C}_{\geq k}| \geq n(\chi+\varepsilon)\}$ and we have the following inequality:

$$\begin{aligned} \mathbb{P}(|\mathbf{C}_{max}| \geq n(\chi+\varepsilon)) &= \mathbb{P}\left(\sum_{v\in[n]} \mathbb{1}_{\{|\mathbf{C}(v)|\geq n(\chi+\varepsilon)\}} \geq n(\chi+\varepsilon)\right) \\ &\leq \mathbb{P}\left(\frac{1}{n}\sum_{v\in[n]} \mathbb{1}_{\{|\mathbf{C}(v)|\geq k\}} \geq (\chi+\varepsilon)\right), \end{aligned} \tag{5.11}$$

where in the last step we choose n so large that $k \leq n(\chi+\varepsilon)$.
Now we know that, $\frac{\mathbf{C}_{\geq k}}{n} \xrightarrow{\mathbb{P}} \mathbb{P}(|G| \geq k)$ which implies that in the limit: $\frac{\mathbf{C}_{\geq k}}{n} \approx \mathbb{P}(|G| \geq k)$. However $\mathbb{P}(|G| \geq k) < \chi + \varepsilon$ hence, $\mathbb{P}\big(\frac{\mathbf{C}_{\geq k}}{n} \geq \chi+\varepsilon\big) \xrightarrow{n\to+\infty} 0$.

Overall, we conclude that:

$$\mathbb{P}(|\mathbf{C}_{max}| \leq n(\chi+\varepsilon)) = 1 - \mathbb{P}(|\mathbf{C}_{max}| \geq n(\chi+\varepsilon)) \xrightarrow{n\to+\infty} 1. \tag{5.12}$$

□

As a result of Corollary 5.1.3, we have that for $\chi = 0$, $\frac{|\mathbf{C}_{max}|}{n} \xrightarrow{\mathbb{P}} 0$, hence a giant component exists only when $\chi > 0$. In the context where the limiting graph is a branching process, we have that the giant component emerges only in the supercritical regime.

5.1.2 Convergence of the Sum of Cluster Sizes

In the following, we may assume that $\chi > 0$. We first define the vector of cluster sizes organised in decreasing order by $(|\mathbf{C}_i|)_{i\geq 1}$. By assumption we have that $|\mathbf{C}_{max}| = |\mathbf{C}_1|$; if we encounter the case where the size of multiple different clusters are equal, then we may choose the label i arbitrarily and still assign their relative size in decreasing order.
Furthermore, we also define a little-o notation in a probability setting by writing that a random variable $X_{n,k} = o_{k,\mathbb{P}}(1)$ when, for all $\varepsilon > 0$,

$$\lim_{k\to+\infty} \limsup_{n\to+\infty} \mathbb{P}(|X_{n,k}| > \varepsilon) = 0 \tag{5.13}$$

We now present two lemmata relating a sum of cluster sizes to the survival probability χ.

Lemma 5.1.4. *(Convergence of sum of cluster sizes)*

$$\frac{\mathbf{C}_{\geq k}}{n} = \chi + o_{k,\mathbb{P}}(1) \tag{5.14}$$

Proof. Following the notation for the vector of cluster sizes, we adjust equation (5.7) to have that:

$$\frac{\mathbf{C}_{\geq k}}{n} = \frac{1}{n}\sum_{v\in[n]} \mathbb{1}_{\{|\mathbf{C}(v)|\geq k\}} = \frac{1}{n}\sum_{i\geq 1} |\mathbf{C}_i| \mathbb{1}_{\{|\mathbf{C}_i|\geq k\}}. \tag{5.15}$$

By the definition of convergence in probability we conclude that, for all $\varepsilon > 0$:

$$\lim_{n\to+\infty} \mathbb{P}\left(\left|\frac{\mathbf{C}_{\geq k}}{n} - \chi_{\geq k}\right| > \varepsilon\right) = 0 \Longrightarrow \limsup_{n\to+\infty} \mathbb{P}\left(\left|\frac{\mathbf{C}_{\geq k}}{n} - \chi_{\geq k}\right| > \varepsilon\right) = 0. \tag{5.16}$$

Now since $\chi_{\geq k} \xrightarrow{k\to+\infty} \chi$, it remains to show that $\frac{\mathbf{C}_{\geq k}}{n} - \chi = o_{k,\mathbb{P}}(1)$ using the definition of the little-o notation introduced in (5.13). We then have the following:

$$\begin{aligned}
\limsup_{n\to+\infty} \mathbb{P}\left(\left|\frac{\mathbf{C}_{\geq k}}{n} - \chi\right| > \varepsilon\right) &= \limsup_{n\to+\infty} \mathbb{P}\left(\left|\frac{\mathbf{C}_{\geq k}}{n} - \chi_{\geq k} + \chi_{\geq k} - \chi\right| > \varepsilon\right) \\
&\leq \limsup_{n\to+\infty} \mathbb{P}\left(\left|\frac{\mathbf{C}_{\geq k}}{n} - \chi_{\geq k}\right| > \varepsilon\right) + \limsup_{n\to+\infty} \mathbb{P}\left(\left|\chi_{\geq k} - \chi\right| > \varepsilon\right) \\
&= \mathbb{P}\left(\left|\chi_{\geq k} - \chi\right| > \varepsilon\right),
\end{aligned} \tag{5.17}$$

where we used (5.16) in the second line of the argument, and the fact that the event $\left\{\left|\chi_{\geq k} - \chi\right| > \varepsilon\right\}$ is independent of n. We can now proceed by taking a limit over k:

$$\lim_{k\to+\infty} \limsup_{n\to+\infty} \mathbb{P}\left(\left|\frac{\mathbf{C}_{\geq k}}{n} - \chi\right| > \varepsilon\right) \leq \lim_{k\to+\infty} \mathbb{P}\left(\left|\chi_{\geq k} - \chi\right| > \varepsilon\right) = 0, \tag{5.18}$$

where we used the fact that $\chi_{\geq k} \xrightarrow{k\to+\infty} \chi$. Overall we have:

$$\lim_{k\to+\infty} \limsup_{n\to+\infty} \mathbb{P}\left(\left|\frac{\mathbf{C}_{\geq k}}{n} - \chi\right| > \varepsilon\right) = 0 \Longrightarrow \frac{\mathbf{C}_{\geq k}}{n} - \chi = o_{k,\mathbb{P}}(1) \Longrightarrow \frac{\mathbf{C}_{\geq k}}{n} = \chi + o_{k,\mathbb{P}}(1), \tag{5.19}$$

which concludes the proof of the lemma. □

We now present a second lemma, similarly presenting a result on the convergence of the sum of squares of cluster sizes.

Lemma 5.1.5. *(Convergence of sum of squares of cluster sizes) Under the conditions of Theorem 5.1.1, the following relation holds:*

$$\frac{\mathbf{C}^2_{\geq k}}{n^2} = \chi^2 + o_{k,\mathbb{P}}(1) \tag{5.20}$$

Proof. By definition,

$$\frac{\mathbf{C}^2_{\geq k}}{n^2} = \frac{1}{n^2}\sum_{i\geq 1} |\mathbf{C}_i|^2 \mathbb{1}_{\{|\mathbf{C}_i|\geq k\}}. \tag{5.21}$$

For this proof we consider a version of the Multinomial Theorem in the case where the power is $n = 2$. It states that for a sequence of real numbers $\{a_i\}_{i=1}^m$ we have:

$$\left(\sum_{i=1}^{m} a_i\right)^2 = \sum_{i=1}^{m} a_i^2 + \sum_{\substack{i,j=1 \\ i\neq j}}^{m} a_i a_j. \tag{5.22}$$

Now, in the context of cluster sizes we can rewrite (5.21) in the following way:

$$\left(\frac{1}{n}\sum_{i\geq 1} |\mathbf{C}_i| \mathbb{1}_{\{|\mathbf{C}_i|\geq k\}}\right)^2 = \frac{1}{n^2}\sum_{i\geq 1} |\mathbf{C}_i|^2 \mathbb{1}_{\{|\mathbf{C}_i|\geq k\}} + \frac{1}{n^2}\sum_{\substack{i,j\geq 1 \\ i\neq j}} |\mathbf{C}_i||\mathbf{C}_j| \mathbb{1}_{\{|\mathbf{C}_i|,|\mathbf{C}_j|\geq k\}}. \tag{5.23}$$

Rearranging the above and using Lemma 5.1.4 we get:

$$\frac{1}{n^2}\sum_{i\geq 1} |\mathbf{C}_i|^2 \mathbb{1}_{\{|\mathbf{C}_i|\geq k\}} = \chi^2 + o_{k,\mathbb{P}}(1) - \frac{1}{n^2}\sum_{\substack{i,j\geq 1 \\ i\neq j}} |\mathbf{C}_i||\mathbf{C}_j| \mathbb{1}_{\{|\mathbf{C}_i|,|\mathbf{C}_j|\geq k\}}. \tag{5.24}$$

It now remains to prove that:

$$\frac{1}{n^2}\sum_{\substack{i,j\geq 1 \\ i\neq j}} |\mathbf{C}_i||\mathbf{C}_j| \mathbb{1}_{\{|\mathbf{C}_i|,|\mathbf{C}_j|\geq k\}} = o_{k,\mathbb{P}}(1). \tag{5.25}$$

For this, we note that the sum $\sum_{\substack{i,j\geq 1 \\ i\neq j}} |\mathbf{C}_i||\mathbf{C}_j| \mathbb{1}_{\{|\mathbf{C}_i|,|\mathbf{C}_j|\geq k\}}$ gives the sum of the product of cluster size for two disjoint clusters, i.e. $\{\mathbf{C}_i \cap \mathbf{C}_j = \varnothing\}$, as we are considering $i \neq j$ along with a product of sizes $|\mathbf{C}_i||\mathbf{C}_j|$ to capture this difference. Furthermore, we grow the size of the clusters by choosing k so large that we end up considering $k \to +\infty$. Hence, writing this event as $\{|\mathbf{C}_i|, |\mathbf{C}_j| \geq k\}$ we have that:

$$\frac{1}{n^2}\sum_{\substack{i,j\geq 1 \\ i\neq j}} |\mathbf{C}_i||\mathbf{C}_j| \mathbb{1}_{\{|\mathbf{C}_i|,|\mathbf{C}_j|\geq k\}} = \frac{1}{n^2}\Big|\{(x,y) \in [n]\times[n] : |\mathbf{C}(x)|, |\mathbf{C}(y)| \geq k, \mathbf{C}(x)\cap\mathbf{C}(y) = \varnothing\}\Big| \tag{5.26}$$

Now using Markov's Inequality and condition (5.4) we have, for all $\varepsilon > 0$,

$$\begin{aligned} &\lim_{k\to+\infty} \limsup_{n\to+\infty} \mathbb{P}\Big(\frac{1}{n^2}\Big|\{(x,y) \in [n]\times[n] : |\mathbf{C}(x)|, |\mathbf{C}(y)| \geq k, \mathbf{C}(x)\cap\mathbf{C}(y) = \varnothing\}\Big| \geq \varepsilon\Big) \\ &\leq \lim_{k\to+\infty} \limsup_{n\to+\infty} \frac{1}{\varepsilon n^2}\mathbb{E}\Big(\Big|\{(x,y) \in [n]\times[n] : |\mathbf{C}(x)|, |\mathbf{C}(y)| \geq k, \mathbf{C}(x)\cap\mathbf{C}(y) = \varnothing\}\Big|\Big) = 0. \end{aligned} \tag{5.27}$$

Overall we conclude that,

$$\lim_{k\to+\infty} \limsup_{n\to+\infty} \mathbb{P}\Big(\frac{1}{n^2}\Big|\{(x,y) \in [n]\times[n] : |\mathbf{C}(x)|, |\mathbf{C}(y)| \geq k, \mathbf{C}(x)\cap\mathbf{C}(y) = \varnothing\}\Big| \geq \varepsilon\Big) = 0. \tag{5.28}$$

Which in turns gives us:

$$\frac{1}{n^2}\sum_{\substack{i,j\geq 1\\ i\neq j}} |\mathbf{C}_i||\mathbf{C}_j|\mathbb{1}_{\{|\mathbf{C}_i|,|\mathbf{C}_j|\geq k\}} = o_{k,\mathbb{P}}(1). \tag{5.29}$$

Finally considering equation (5.24) we get the result,

$$\frac{1}{n^2}\sum_{i\geq 1}|\mathbf{C}_i|^2\mathbb{1}_{\{|\mathbf{C}_i|\geq k\}} = \chi^2 + o_{k,\mathbb{P}}(1) - \frac{1}{n^2}\sum_{\substack{i,j\geq 1\\ i\neq j}} |\mathbf{C}_i||\mathbf{C}_j|\mathbb{1}_{\{|\mathbf{C}_i|,|\mathbf{C}_j|\geq k\}} = \chi^2 + o_{k,\mathbb{P}}(1). \tag{5.30}$$

□

5.1.3 Proof of "The Giant is Almost Local"

In this section, we aim to prove Theorem 5.1.1 using the theory presented in the previous two sections. In this proof we use the fact that, for a sequence $\{a_i\}_{i=1}^m$ of positive real numbers,

$$\sum_{i=1}^m a_i^2 \leq \Big(\sum_{i=1}^m a_i\Big)^2. \tag{5.31}$$

Recall that we aim to show that in the limit, $|\mathbf{C}_{max}| \approx n\chi$. Now we know that for all $i \geq 1$, $|\mathbf{C}_{max}| \geq |\mathbf{C}_i|$. But in fact, we can generalise this statement by also noting that $|\mathbf{C}_{max}| > \sum_{i\geq 2}|\mathbf{C}_i|$. Hence by Lemma 5.1.5, and using the fact that for all $i \geq 1$, $|\mathbf{C}_{max}| \geq |\mathbf{C}_i|$:

$$\begin{aligned}\chi^2 + o_{k,\mathbb{P}}(1) = \frac{1}{n^2}\sum_{i\geq 1}|\mathbf{C}_i|^2\mathbb{1}_{\{|\mathbf{C}_i|\geq k\}} &\leq \frac{1}{n^2}\Big(\sum_{i\geq 1}|\mathbf{C}_i|\mathbb{1}_{\{|\mathbf{C}_i|\geq k\}}\Big)^2 \\ &\leq \frac{|\mathbf{C}_{max}|}{n}\Big(\sum_{i\geq 1}|\mathbf{C}_i|\mathbb{1}_{\{|\mathbf{C}_i|\geq k\}}\Big),\end{aligned} \tag{5.32}$$

where the last step holds from (5.31). Hence, using Lemma 5.1.4,

$$\chi^2 + o_{k,\mathbb{P}}(1) \leq \frac{|\mathbf{C}_{max}|}{n}(\chi + o_{k,\mathbb{P}}(1)). \tag{5.33}$$

Rearranging this expression, we get:

$$\frac{|\mathbf{C}_{max}|}{n} \geq \frac{\chi^2 + o_{k,\mathbb{P}}(1)}{\chi + o_{k,\mathbb{P}}(1)} = \frac{(\chi + o_{k,\mathbb{P}}(1))^2}{\chi + o_{k,\mathbb{P}}(1)} = \chi + o_{k,\mathbb{P}}(1). \tag{5.34}$$

The last relation holds since for any constant $c \in \mathbb{R}_{>0}$ and for any random variable $X_{n,k} = o_{k,\mathbb{P}}(1)$ we have: $cX_{n,k} = o_{k,\mathbb{P}}(1)$.
Now we know that $|\mathbf{C}_{max}| \geq n(\chi + o_{k,\mathbb{P}}(1))$ and since we choose k arbitrarily we can reduce the little-o notation to $|\mathbf{C}_{max}| \geq n(\chi + o_{\mathbb{P}}(1))$, where for a random variable X_n and a strictly positive sequence of real numbers a_n, $X_n = o_{\mathbb{P}}(a_n) \Longrightarrow \frac{X_n}{a_n} \xrightarrow{\mathbb{P}} 0$, as described in [14].
We also know, by Corollary 5.1.3, that $|\mathbf{C}_{max}| \leq n(\chi + o_{\mathbb{P}}(1))$. So in fact we must have that $|\mathbf{C}_{max}| =$

$n(\chi + o_{\mathbb{P}}(1))$ which proves that:

$$\frac{|\mathbf{C}_{max}|}{n} \xrightarrow{\mathbb{P}} \chi. \tag{5.35}$$

It now remains to prove that $\frac{|\mathbf{C}_2|}{n} \xrightarrow{\mathbb{P}} 0$ where $\mathbf{C}_2$ is the second largest component. In a similar way to proving the proportion of vertices in $\mathbf{C}_{max}$, we condition the size of $\mathbf{C}_2$ to be $|\mathbf{C}_2| \geq k$, for all $k \geq 1$. Therefore,

$$\begin{aligned} \frac{|\mathbf{C}_2|}{n} \leq \frac{1}{n}\sum_{i\geq 2} |\mathbf{C}_i| \mathbb{1}_{\{|\mathbf{C}_i|\geq k\}} &= \frac{1}{n}\sum_{i\geq 1} |\mathbf{C}_i| \mathbb{1}_{\{|\mathbf{C}_i|\geq k\}} - \frac{|\mathbf{C}_{max}|}{n} \\ &= \frac{\mathbf{C}_{\geq k}}{n} - \frac{|\mathbf{C}_{max}|}{n}. \end{aligned} \tag{5.36}$$

Now using Lemma 5.1.4 we have,

$$\frac{|\mathbf{C}_2|}{n} \leq \chi + o_{k,\mathbb{P}}(1) - (\chi + o_{\mathbb{P}}(1)) = o_{k,\mathbb{P}}(1) - o_{\mathbb{P}}(1). \tag{5.37}$$

And again, since k is arbitrary, we have the required result:

$$\frac{|\mathbf{C}_2|}{n} \leq o_{\mathbb{P}}(1) \Longrightarrow \frac{|\mathbf{C}_2|}{n} \xrightarrow{\mathbb{P}} 0. \tag{5.38}$$

This concludes the proof of the Theorem.

Remark 5.1.6. $\left(|\mathbf{C}_{max}| > \sum_{i\geq 2} |\mathbf{C}_i| \text{ for } ER_n(\frac{\lambda}{n})\right)$
In the case of an Erdős-Rényi random graph (we will study this model in the next section) we have that, excluding $\mathbf{C}_{max}$ which has size $O(n)$, all other components have size at most $O(\log(n))$. Hence it holds that for all $i \geq 1$, $|\mathbf{C}_{max}| \geq |\mathbf{C}_i|$. But in fact, we can generalise this statement by also noting that $\sum_{i\geq 2} |\mathbf{C}_i| = O(\log(n))$. And since powers beat logarithms we conclude that the size of the giant component is greater than the sum of all other cluster sizes, i.e. $|\mathbf{C}_{max}| > \sum_{i\geq 2} |\mathbf{C}_i|$.

5.2 An Application to the Erdős-Rényi Random Graph

We now end this section on an application to an Erdős-Rényi Random Graph $ER_n(\frac{\lambda}{n})$. We will first present the exploration process of the greatest component in this graph before moving on to the study of its size. For this, we will briefly describe the "old" approach mainly using advanced combinatorial arguments, and then present a new version using local limits based on the previous section.

5.2.1 Exploration of the Giant Component

First, for a vertex v for which we have uncovered its neighbours, we say that v is *saturated* and that its neighbours are *explored*. Now, recall that in the Erdős-Rényi random graph edges appear independently where the degrees follow a Binomial distribution $Bin(n-1, \frac{\lambda}{n})$. We now explain the breadth-first exploration of the graph for the detection of the giant component, where we label the vertices by $v_k \in [n]$ for $k \in \{1, 2, ..., n\}$. In this type of exploration we inspect each node individually

and identify all the neighbours before moving on to another vertex. First pick a node v_1 in $ER_n(\frac{\lambda}{n})$ at random and find all of its neighbours $\{v_2,...,v_r\}$ where $r \leq n$. Once that has been done, we move on to another vertex, say v_2, and find its neighbours from the vertex set $[n]\backslash\{v_3,...,v_r\}$. We continue this procedure until all vertices in the graph have been explored, and at every point in the search labeling the saturated vertices once we have found its neighbours.

We can also remark the following: at the beginning of the search, if v_1 has a number of neighbours equal to S_1 then, as we know, this number follows a Binomial random variable $Bin(n-1,\frac{\lambda}{n})$. If we now look at the number S_2 of neighbours of v_2, we would have that this number follows a Binomial random variable $Bin(n-1-S_1,\frac{\lambda}{n})$. Iterating this process at each step in the search for the neighbourhood of a vertex v_r, for $r \leq n$, we have that the number of neighbours in the next generation follows a Binomial distribution $S_r \sim Bin(n-1-\sum_{i=1}^{r-1} S_i,\frac{\lambda}{n})$.

5.2.2 Old Approach to the Problem

In the paper written in 1960 by Erdős and Rényi [1], it is shown that for $\lambda < 1$, the size of the largest component is almost surely at most $O(\log(n))$. Furthermore, at the critical point $\lambda = 1$, one can almost surely find a large component of size at most $n^{\frac{2}{3}}$. However, in the case where $\lambda > 1$, a high level of clustering appears with the emergence of a so called giant component, where we can almost surely find a significant number of vertices in this single component. The theorem for the supercritical case is the following:

Theorem 5.2.1. *For $\lambda > 1$, there is a unique giant component of size $n(\chi_\lambda + o_{\mathbb{P}}(1))$, where χ_λ is the survival probability of a Poisson branching process. In addition, all other components are almost surely of size at most $O(\log(n))$ as $n \to +\infty$.*

The proof of this theorem revolves around the usage of rather advanced combinatorial arguments, to which we will present the main steps and ideas. This summary is based on two sets of lecture notes [15], [16] where the full details can be found in an elaborate paper written by Svante Janson, Tomasz Luczak and Andrzej Rucinski [17].

Sketch of Proof. We first define some notation for the two different sizes at the threshold of the branching process, setting $k_- = \alpha\log(n)$ and $k_+ = n^{\frac{2}{3}}$ for some constant α that will be determined later. The idea of the proof is to proceed by a sectioning argument, and show that for different sizes of components, i.e. for different sets of values of k, the only possibility for the size of the unique giant component is $n(\chi_\lambda + o_{\mathbb{P}}(1))$.

Step 1: We identify the possible ranges for the size of the largest and second largest connected components. The argument resides in proving the following claim: For $k_- \leq k \leq k_+$ the branching process either terminates in fewer than k_- steps or continues for more than k_+ steps. In other words, there is no connected component whose size lies in the range $k_- \leq k \leq k_+$. This claim allows us to conclude that either vertices are part of a component whose size is at most $O(\log(n))$ or belongs to a large component whose size is at least $O(n^{\frac{2}{3}})$.

We note that it is at this step that the constant α is chosen so that we can bound above the probability to find a vertex in a component of size k where $k_- \leq k \leq k_+$. Choosing $\alpha = \frac{16\lambda}{(\lambda-1)^2}$ guarantees this upper bound.

Step 2: The next task is to show that as $n \to +\infty$, there almost surely exists a unique component containing all the vertices for which the branching process survives for at least $O(n^{\frac{2}{3}})$ steps. One can prove this fact by taking two distinct vertices u and v at random in the graph and run the two branching processes independently to at least k_+ steps. Of course if we find a vertex in common in the first k_+ steps there is nothing to prove. However, it can be shown that for any pair of vertices (u,v) in the graph, running two branching processes beyond k_+ steps from u and v, the probability to not find an edge connecting the component around u and the component around v is $o(1)$. Therefore, it almost surely guarantees the uniqueness of a component containing all the vertices for which the branching process survives for at least $O(n^{\frac{2}{3}})$ steps.

Step 3: From now on, we define Γ to be the set of all vertices v for which the branching process from v ends in at most k_- steps. Now, the final step of the proof is to show that the number of vertices belonging to Γ in the process is $(1-\chi_\lambda+o(1))n$. This can be shown by a direct application of Chebyshev's Inequality. By defining the random variable Z to be $Z=\sum_v Z_v$, where $Z_v = 1$ if $v \in \Gamma$ and $Z_v = 0$ otherwise, it is clear that Z represents the number of vertices outside of the giant component. Now, one may show that $\mathbb{P}(v \in \Gamma) \xrightarrow{n\to+\infty} 1-\chi_\lambda$ and as a consequence $\mathbb{E}(Z_v) = \mathbb{P}(v \in \Gamma) \xrightarrow{n\to+\infty} 1-\chi_\lambda$. Hence $\mathbb{E}(Z) = (1-\chi_\lambda+o(1))n$ and finally showing that $\mathbb{E}(Z^2) \leq \mathbb{E}(Z)^2(1+o(1))$ we conclude using Chebyshev's Inequality that, in the limit, Γ contains $(1-\chi_\lambda+o(1))n$. Therefore the number of vertices in the giant component is $n-(1-\chi_\lambda+o_{\mathbb{P}}(1))n = n(\chi_\lambda+o_{\mathbb{P}}(1))$. □

5.2.3 New Approach Using Local Limits

In this section, we aim to present the new perspective behind a local limit approach to the giant component in an Erdős-Rényi random graph. For the purpose of the main theorem, we state without proof the following result on the local limit of $ER_n(\frac{\lambda}{n})$ [8, pp.60]:

Theorem 5.2.2. $\big($*Local limit of* $ER_n(\frac{\lambda}{n})\big)$ *Fix* $\lambda > 0$. $ER_n(\frac{\lambda}{n})$ *converges locally in probability to a Poisson branching process with mean offspring* λ.

We note that this result may seem unsurprising, assuming that locally-tree like structure in random graphs can be expected. Recall that for the Erdős-Rényi random graph, edges appear independently following a Binomial random variable $Bin(n-1,\frac{\lambda}{n})$. Then if we perform an exploration process and explore α vertices in the graph, then the creation of the upcoming edge will follow a Binomial $Bin(n-\alpha,\frac{\lambda}{n})$. If we now assume that the number of vertices gets large enough, we may use the well-known Poisson approximation to the Binomial distribution to find that, as $n \to +\infty$, $Bin(n-\alpha,\frac{\lambda}{n}) \xrightarrow{d} Poi(\lambda)$. This argument explains the particular type of offspring distribution for the local limit of $ER_n(\frac{\lambda}{n})$.

Theorem 5.2.3. *[8, pp.80] Fix $\lambda > 0$ and denote the boundary of the neighbourhood of a vertex ρ by $\partial B_r^{(G_n)}(\rho)$. Under the assumptions of Theorem 5.1.1 consider the following condition:*

$$\lim_{r\to+\infty} \limsup_{n\to+\infty} \frac{1}{n^2}\mathbb{E}\left[|\{(x,y)\in[n]\times[n] : |\partial B_r^{(G_n)}(x)|, |\partial B_r^{(G_n)}(y)| \geq r, \partial B_r^{(G_n)}(x)\cap\partial B_r^{(G_n)}(y) = \varnothing\}|\right] = 0, \tag{5.39}$$

provided that there exists $r \to +\infty$ dependent on k such that,

$$\mathbb{P}(|\mathbf{C}(\rho)| \geq k, |\partial B_r^{(G_n)}(\rho)| < r) \xrightarrow{k\to+\infty} 0 \quad \textit{and,} \quad \mathbb{P}(|\mathbf{C}(\rho)| < k, |\partial B_r^{(G_n)}(\rho)| \geq r) \xrightarrow{k\to+\infty} 0. \tag{5.40}$$

Now recall that χ_λ is the survival probability of a Poisson branching process with mean offspring distribution λ. Then:

$$\frac{|\mathbf{C}_{max}|}{n} \xrightarrow{\mathbb{P}} \chi_\lambda \qquad \textit{and,} \qquad \frac{|\mathbf{C}_2|}{n} \xrightarrow{\mathbb{P}} 0. \tag{5.41}$$

In contrast to the approach in subsection 5.2.2, this theorem does not give any information on the potential size of the second largest component, whereas Theorem 5.2.1 asserts that for all $i \geq 2$, $|\mathbf{C}_i| = O(\log(n))$.

For the purpose of this proof we will use a result related to Weierstrass product inequalities [18, pp.1] which we present in the following form:

Lemma 5.2.4. *Consider $0 \leq p_n < 1$. If $\sum_{n=1}^{\infty} p_n \leq 1$, then:*

$$\prod_{n=1}^{\infty}(1-p_n) \geq 1 - \sum_{n=1}^{\infty} p_n \tag{5.42}$$

Proof. Of course, if $0 \leq p_n < 1$ then $0 < 1 - p_n \leq 1$.

Now, suppose first that $\sum_{n=1}^{\infty} p_n = 1$ then it is clear that $\prod_{n=1}^{\infty}(1-p_n) \geq 1 - \sum_{n=1}^{\infty} p_n = 0$.

Suppose now that $\sum_{n=1}^{\infty} p_n < 1$. We will prove (5.42) by induction on n.

Clearly, for $n = 1$, $1 - p_1 \geq 1 - p_1$ which initialises the induction.

Consider now that $\prod_{k=1}^{n}(1-p_k) \geq 1 - \sum_{k=1}^{n} p_k$ holds for the first n terms. We will now show that it still holds for the term $(n+1)$. For this we will freely use that for $a, b \geq 0$, we have $(1-a)(1-b) \geq 1-a-b$.

$$\begin{aligned}\prod_{k=1}^{n+1}(1-p_k) = \prod_{k=1}^{n}(1-p_k)(1-p_{n+1}) &\geq \left(1 - \sum_{k=1}^{n} p_k\right)(1-p_{n+1}) \\ &\geq 1 - \sum_{k=1}^{n} p_k - p_{n+1} = 1 - \sum_{k=1}^{n+1} p_k\end{aligned} \tag{5.43}$$

Then taking $n \to +\infty$, we get the required result. □

We will also use without proof a result on the tails of a Binomial random variable, adapted from well-known bounds on Binomials that can be found in [17, pp.26],

Lemma 5.2.5. *Let $X \sim Bin(n,p)$. Then, for every $\delta > 0$,*

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq \delta\mathbb{E}(X)) \leq 2\exp\left(-\frac{\delta^2\mathbb{E}(X)}{2(1+\delta/3)}\right). \tag{5.44}$$

And we will finally present without proof a chain rule in the probability setting that will be useful when considering an intersection of events:

Lemma 5.2.6. *(Chain rule for probabilities) For events $A_1, A_2, ..., A_n$ whose intersection does not have probability zero, the chain rule states that:*

$$\mathbb{P}\left(\bigcap_{k=1}^{n} A_k\right) = \prod_{k=1}^{n} \mathbb{P}\left(A_k \middle| \bigcap_{i=1}^{k-1} A_i\right). \tag{5.45}$$

Proof. The following steps of the proof are from [8], however we explain the arguments and derivations in a lot more depth than in the second volume of Random Graphs and Complex Networks. We also provide two figures made on Geogebra to clarify the overall idea and key step of the proof. We first observe that, considering Theorem 5.1.1 on the Giant is almost local, the result will follow immediately if condition (5.4) holds. However, we are considering a slightly different but sufficient condition, where instead of growing the size of the whole neighbourhood we are only growing its boundary. Hence this proof will consist of showing that (5.39) holds.
The key idea is to grow the boundaries around two random roots (ρ_1, ρ_2) and find a way to give a good control over the number of vertices in these boundaries. Such control would give us the capacity to manage the number of vertices that have not yet been used.
We can remark that Lemma 5.2.5 guaranties that for n and r large enough, $|\partial B_{r+1}(\rho_1)| \approx \lambda |\partial B_r(\rho_1)|$. The natural idea would be to generalise this result to an arbitrary radius, and we will in fact show that with high probability and in the limit, $|\partial B_{r+k}(\rho_1)| \approx \lambda^k |\partial B_r(\rho_1)|$. This will give us the main ingredient to study the growth. As a result, in approximately $a \log_\lambda(n)$ steps, the boundary will have expanded to n^a vertices.
We will then be able to prove (5.39) by growing the neighbourhoods of $B_r(\rho_1)$ and $B_r(\rho_2)$. Indeed, we will describe the process of growth by first growing $\partial B_{r+k}(\rho_1)$ for $k \geq 1$ until it is very large, and then grow $\partial B_{r+k}(\rho_2)$ for $k \geq r$ until it is also very large. Then, we will show that with a high probability, there is an edge connecting the two resulting boundaries.

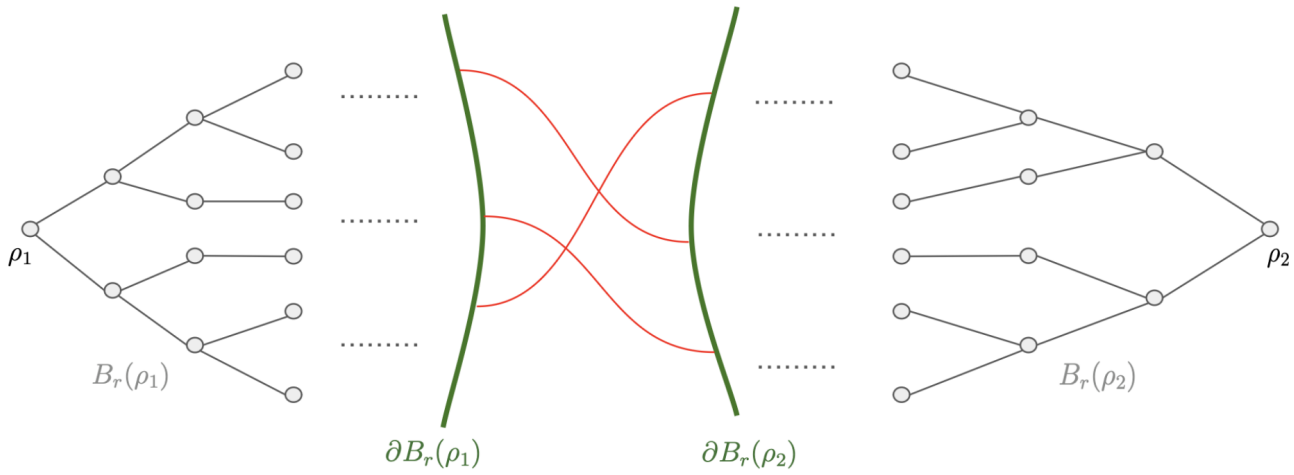


Figure 5.1: Diagram representing the idea behind the proof. We grow the neighbourhood of the graph around ρ_1 and ρ_2 (in green), and with high probability we will find at least one edge connecting the two resulting boundaries (in red). [This figure has been created using GeoGebra(2024)]

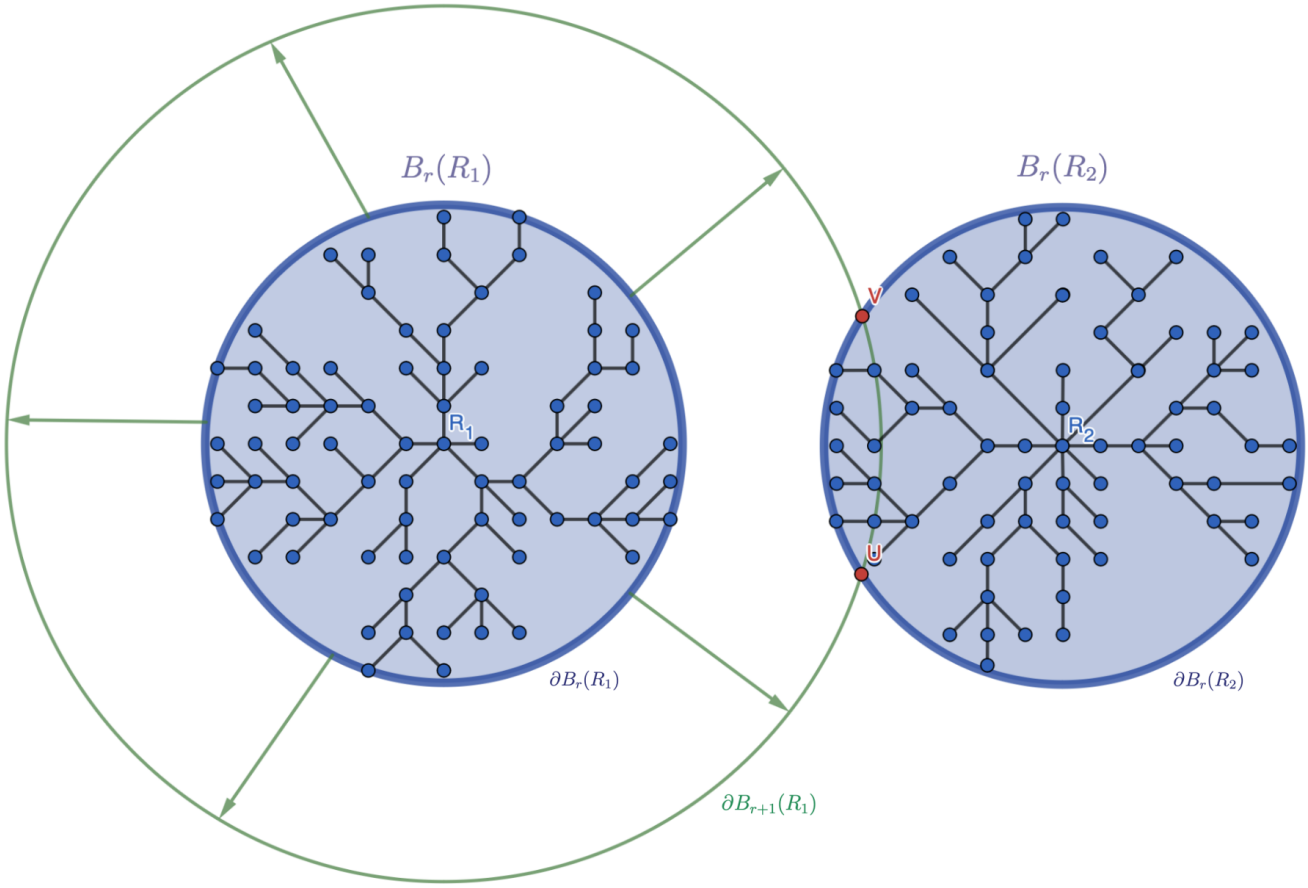


Figure 5.2: The above figure shows the process of growing $\partial B_r(R_1)$ to $\partial B_{r+1}(R_1)$. In this representation, the sector between U and V represents the overlap between $\partial B_{r+1}(R_1)$ and $B_r(R_2)$. Since we want to grow $\partial B_r(R_1)$ outside of $B_r(R_2)$ we will have to adjust the parameters for the creation of edges in the boundary. [This figure has been created using GeoGebra(2024)]

Step 1: Setup

As a matter of clarity, we simplify the notation introduced above by defining $B_r^{(G_n)}(\rho_1) = B_r(\rho_1)$ and $\partial B_r^{(G_n)}(\rho_1) = \partial B_r(\rho_1)$. We fix $G_n = ER_n(\frac{\lambda}{n})$ where $\lambda > 1$.

Since we choose (ρ_1, ρ_2) uniformly at random in the graph, (5.39) can be viewed as:

$$\begin{aligned}\frac{1}{n^2}\mathbb{E}\big[|\{(x,y) \in [n] \times [n] : |\partial B_r(x)|, |\partial B_r(y)| \geq r, \partial B_r(x) \cap \partial B_r(y) = \varnothing\}|\big] \\ = \mathbb{P}(|\partial B_r(\rho_1)|, |\partial B_r(\rho_2)| \geq r, \partial B_r(\rho_1) \cap \partial B_r(\rho_2) = \varnothing).\end{aligned} \tag{5.46}$$

For this proof, we condition $ER_n(\frac{\lambda}{n})$ on the size of the neighbourhoods and its boundaries by $|B_r(\rho_i)| = \alpha_0^i$ and $|\partial B_r(\rho_i)| = \beta_0^i$ for $i \in \{1,2\}$. Then using the Law of Total Probability we can rewrite (5.46) to have:

$$\begin{aligned}&\mathbb{P}(|\partial B_r(\rho_1)|, |\partial B_r(\rho_2)| \geq r, \partial B_r(\rho_1) \cap \partial B_r(\rho_2) = \varnothing) \\ &= \sum_{i \in \{1,2\}} \mathbb{P}\big(|\partial B_r(\rho_1)|, |\partial B_r(\rho_2)| \geq r, \partial B_r(\rho_1) \cap \partial B_r(\rho_2) = \varnothing \big| |B_r(\rho_i)| = \alpha_0^i, |\partial B_r(\rho_i)| = \beta_0^i\big) \\ &\qquad \times \mathbb{P}(|B_r(\rho_i)| = \alpha_0^i, |\partial B_r(\rho_i)| = \beta_0^i).\end{aligned} \tag{5.47}$$

And if we now set $X_{n,r} = \{|\partial B_r(\rho_1)|, |\partial B_r(\rho_2)| \geq r, \partial B_r(\rho_1) \cap \partial B_r(\rho_2) = \varnothing\}$ and $\mathbf{B}_r^i = \{|B_r(\rho_i)| = \alpha_0^i, |\partial B_r(\rho_i)| = \beta_0^i\}$ then (5.47) becomes:

$$\mathbb{P}(X_{n,r}) = \sum_{i \in \{1,2\}} \mathbb{P}(X_{n,r} | \mathbf{B}_r^i) \mathbb{P}(\mathbf{B}_r^i). \tag{5.48}$$

Now our aim is to show that, under condition (5.39), $\mathbb{P}(X_{n,r}) = o(1)$. Since $\mathbb{P}(\mathbf{B}_r^i)$ is fixed, we need to find a radius $r = r(\varepsilon)$ such that for every $\varepsilon > 0$ and for every $\beta_0^1, \beta_0^2 \geq r$ and α_0^1, α_0^2 fixed,

$$\mathbb{P}(X_{n,r} | \mathbf{B}_r^i) = o(1) \Longrightarrow \limsup_{n \to +\infty} \mathbb{P}(X_{n,r} | \mathbf{B}_r^i) \leq \varepsilon. \tag{5.49}$$

Under the given conditioning, $|\partial B_{r+1}(\rho_1) \backslash B_r(\rho_2)| \sim Bin(n_1^1, p_1^1)$, where $n_1^1 = n - \alpha_0^1 - \alpha_0^2$ and $p_1^1 = 1 - (1 - \frac{\lambda}{n})^{\beta_0^1}$. Note that p_1^1 represents the probability of creating $|\partial B_r(\rho_1)| = \beta_0^1$ edges.

Step 2: Growth of $\partial B_r(\rho_1)$

First recall that we want to grow $\partial B_{r+k-1}(\rho_1)$ outside of $B_r(\rho_2)$. We will then write $|\partial B_{r+k}^\dagger(\rho_1)|$ to denote the size of the resulting neighbourhood, reflecting the change in the number of vertices.

In this step, we will make a small ε-variation in the estimate of the size of the neighbourhood. Wanting to prove that in the limit $|\partial B_{r+k}(\rho_1)| \approx \lambda^k |\partial B_r(\rho_1)|$, we consider a small change and define $\overline{\beta_k^1} = \beta_0^1 [\lambda(1+\varepsilon)]^k$ and $\underline{\beta_k^1} = \beta_0^1 [\lambda(1-\varepsilon)]^k$ as upper and lower bounds on $|\partial B_{r+k}^\dagger(\rho_1)|$. Then our aim is to show that with high probability:

$$\underline{\beta_k^1} \leq |\partial B_{r+k}^\dagger(\rho_1)| \leq \overline{\beta_k^1}. \tag{5.50}$$

For the above to hold, we make sure to take $\varepsilon > 0$ small so that $\lambda(1-\varepsilon) > 1$ and $\lambda(1+\varepsilon) > 1$.

We now wish to find the distribution of $|\partial B_{r+k}^\dagger(\rho_1)|$. For this, we will find an upper bound on $|\partial B_{r+k-1}^\dagger(\rho_1)|$ so that under a given conditioning we can deduce a size following a Binomial distribution.

Naturally, this upper bound can be seen as summing the number of vertices already in $B_r(\rho_1)$ along with the number of new vertices when we grow the neighbourhood of $B_r(\rho_1)$. Denoting this upper bound by $\overline{\alpha_{k-1}^1}$, we find:

$$\begin{aligned} |\partial B_{r+k-1}^\dagger(\rho_1)| \leq \overline{\alpha_{k-1}^1} = \alpha_0^1 + \sum_{u=0}^{k-1} \overline{\beta_u^1} &= \alpha_0^1 + \beta_0^1 \sum_{u=0}^{k-1} [\lambda(1+\varepsilon)]^u \\ &= \alpha_0^1 + \beta_0^1 \times \frac{[\lambda(1+\varepsilon)]^k - 1}{\lambda(1+\varepsilon) - 1} \\ &\leq \alpha_0^1 + \beta_0^1 \times \frac{[\lambda(1+\varepsilon)]^k}{\lambda(1+\varepsilon) - 1}. \end{aligned} \tag{5.51}$$

As explained in the beginning of this proof, we consider that the boundary will have expanded to n^a vertices in $\lceil a \log_\lambda(n) \rceil$ steps. Then, still with the ε-variation, and for $a \in (\frac{1}{2}, 1)$, we may consider

$k \leq k_n^* = \lceil a \log_{\lambda(1-\varepsilon)}(n) \rceil$. Therefore,

$$|\partial B^{\dagger}_{r+k-1}(\rho_1)| \leq \alpha_0^1 + \beta_0^1 \times \frac{[\lambda(1+\varepsilon)]^{a\log_{\lambda(1-\varepsilon)}(n)}}{\lambda(1+\varepsilon)-1}. \tag{5.52}$$

Furthermore, we can change the base of the logarithm to the natural logarithm to find:

$$a\log_{\lambda(1-\varepsilon)}(n) = \log_{\lambda(1-\varepsilon)}(n^a) = \frac{\log(n^a)}{\log(\lambda(1-\varepsilon))}. \tag{5.53}$$

Hence we can rewrite the numerator in (5.52) as:

$$\begin{aligned}
[\lambda(1+\varepsilon)]^{\log_{\lambda(1-\varepsilon)}(n^a)} = e^{\log\left([\lambda(1+\varepsilon)]^{\log_{\lambda(1-\varepsilon)}(n^a)}\right)} &= e^{\log(\lambda(1+\varepsilon))\log_{\lambda(1-\varepsilon)}(n^a)} \\
&= e^{\log(\lambda(1+\varepsilon))\frac{\log(n^a)}{\log(\lambda(1-\varepsilon))}} \\
&= n^{a\frac{\log(\lambda(1+\varepsilon))}{\log(\lambda(1-\varepsilon))}}.
\end{aligned} \tag{5.54}$$

Therefore, considering (5.52), we can find a constant $C > 1$ such that,

$$|\partial B^{\dagger}_{r+k-1}(\rho_1)| \leq \alpha_0^1 + \frac{\beta_0^1}{(\lambda(1+\varepsilon))-1} n^{a\frac{\log(\lambda(1+\varepsilon))}{\log(\lambda(1-\varepsilon))}} \leq C n^{a\frac{\log(\lambda(1+\varepsilon))}{\log(\lambda(1-\varepsilon))}}. \tag{5.55}$$

In the above inequality we will choose $a \in (\frac{1}{2}, 1)$ so that $a\frac{\log(\lambda(1+\varepsilon))}{\log(\lambda(1-\varepsilon))} \in (\frac{1}{2}, 1)$.
Now if we condition on $|\partial B^{\dagger}_{r+k-1}(\rho_1)| = \beta_{k-1}^1 \in \left[\underline{\beta_{k-1}^1}, \overline{\beta_{k-1}^1}\right]$ and $|B^{\dagger}_{r+k-1}(\rho_1)| = \alpha_{k-1}^1 \leq \overline{\alpha_{k-1}^1}$ we have the following aim result on the distribution of the size of $\partial B^{\dagger}_{r+k}(\rho_1)$:

$$|\partial B^{\dagger}_{r+k}(\rho_1)| \sim Bin(n_k^1, p_k^1), \tag{5.56}$$

where we need to reflect in the parameters the fact that we grow $\partial B_{r+k-1}(\rho_1)$ outside of $B_r(\rho_2)$. This can be done by subtracting α_0^2 from the number of available vertices. Hence the coefficients of the Binomial random variable are:

$$n_k^1 = n - \alpha_{k-1}^1 - \alpha_0^2 \qquad \text{and,} \qquad p_k^1 = 1 - \left(1 - \frac{\lambda}{n}\right)^{\beta_{k-1}^1}. \tag{5.57}$$

Now that we know how the size of $\partial B^{\dagger}_{r+k}(\rho_1)$ behaves, we can finally study the event:

$$\Gamma_{r,k}^1 = \left\{\underline{\beta_k^1} \leq |\partial B^{\dagger}_{r+k}(\rho_1)| \leq \overline{\beta_k^1}\right\}. \tag{5.58}$$

The aim of the study is to find the probability of the following general event:

$$\Gamma_{r,[k]}^1 = \bigcap_{u\in[k]} \Gamma_{r,u}^1 = \bigcap_{u\in[k]} \left\{\underline{\beta_u^1} \leq |\partial B^{\dagger}_{r+u}(\rho_1)| \leq \overline{\beta_u^1}\right\}. \tag{5.59}$$

Now recall that we want to show the above result with high probability under the conditions given in the setup under $\mathbf{B}_r^i$. To again simplify the notation, we write $\mathbb{P}_{\mathbf{B}_r^i}$ and $\mathbb{E}_{\mathbf{B}_r^i}$ to be the probability and the expectation under the conditioning $\mathbf{B}_r^i$. In what follows, we aim to find a lower bound on $\mathbb{P}_{\mathbf{B}_r^i}$.
We have that, using the chain rule from Lemma 5.2.6:

$$\begin{aligned}\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,[k]}\right) = \mathbb{P}_{\mathbf{B}_r^i}\left(\bigcap_{u\in[k]}\Gamma^1_{r,u}\right) &= \prod_{u\in[k]}\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,u}\middle|\bigcap_{j\in[u-1]}\Gamma^1_{r,j}\right)\\ &= \prod_{u\in[k]}\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,u}\middle|\Gamma^1_{r,[u-1]}\right).\end{aligned} \tag{5.60}$$

Furthermore, since for two general events A and B, $\mathbb{P}(A|B) = 1-\mathbb{P}(A^c|B)$, we can apply Lemma 5.2.4 to find:

$$\begin{aligned}\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,[k]}\right) = \prod_{u\in[k]}\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,u}\middle|\Gamma^1_{r,[u-1]}\right) &= \prod_{u\in[k]}\left(1-\mathbb{P}_{\mathbf{B}_r^i}\left((\Gamma^1_{r,u})^c\middle|\Gamma^1_{r,[u-1]}\right)\right)\\ &\geq 1-\sum_{u\in[k]}\mathbb{P}_{\mathbf{B}_r^i}\left((\Gamma^1_{r,u})^c\middle|\Gamma^1_{r,[u-1]}\right).\end{aligned} \tag{5.61}$$

It now remains to simplify, or at least rewrite the conditional probability in the last line to a simpler expression. To do this, we will use Lemma 5.2.5 to $|\partial B^\dagger_{r+k}(\rho_1)|$ along with $\delta = \varepsilon/2$ on its expectation, where we have that $\mathbb{E}_{\mathbf{B}_r^i}\left(|\partial B^\dagger_{r+k}(\rho_1)|\middle|\Gamma^1_{r,[u-1]}\right) = n^1_k p^1_k$. We need to provide some upper and lower bounds on $n^1_k p^1_k$, where we first provide bounds on p^1_k. For the upper bound, we use a version of Bernoulli's inequality that can be found in [19] stating that for $t \geq 1$ and $0 \leq x \leq 1$ we have that $(1-x)^t \geq 1-xt$. Then by the definition of p^1_k given in (5.57) we have:

$$\left(1-\frac{\lambda}{n}\right)^{\beta^1_{k-1}} \geq 1-\frac{\lambda}{n}\beta^1_{k-1} \Longleftrightarrow p^1_k \leq \frac{\lambda}{n}\beta^1_{k-1}. \tag{5.62}$$

For the lower bound now, we use an inequality related to Bernoulli's inequality, and it states that for any $x \in \mathbb{R}$ and $r > 0$ one has $(1+x)^r \leq e^{xr}$. Then using again the definition of p^1_k, we obtain:

$$\left(1-\frac{\lambda}{n}\right)^{\beta^1_{k-1}} \leq e^{-\frac{\lambda}{n}\beta^1_{k-1}} = \sum_{k=0}^{\infty}\frac{(-\frac{\lambda}{n}\beta^1_{k-1})^k}{k!} = 1-\frac{\beta^1_{k-1}\lambda}{n}+\frac{(\beta^1_{k-1}\lambda)^2}{2n^2}+O\left(\frac{1}{n^3}\right). \tag{5.63}$$

This implies that we have the following lower bound for p^1_k:

$$p^1_k \geq \frac{\beta^1_{k-1}\lambda}{n}-\frac{(\beta^1_{k-1}\lambda)^2}{2n^2}+O\left(\frac{1}{n^3}\right) \geq \frac{\beta^1_{k-1}\lambda}{n}-\frac{(\beta^1_{k-1}\lambda)^2}{2n^2}. \tag{5.64}$$

Hence we get the following bounds for p^1_k:

$$\frac{\beta^1_{k-1}\lambda}{n}-\frac{(\beta^1_{k-1}\lambda)^2}{2n^2} \leq p^1_k \leq \frac{\beta^1_{k-1}\lambda}{n}. \tag{5.65}$$

We are now ready to give bounds on $n_k^1 p_k^1$. If we consider n to be sufficiently large, we may bound p_k^1 and n_k^1 by our small ε-variation to obtain:

$$p_k^1 \geq \frac{\beta_{k-1}^1 \lambda}{n}\left(1-\frac{\beta_{k-1}^1\lambda}{2n}\right) \geq \frac{\beta_{k-1}^1\lambda}{n}\left(1-\frac{\varepsilon}{4}\right), \tag{5.66}$$

along with,

$$n_k^1 = n-\alpha_{k-1}^1-\alpha_0^2 = n\left(1-\frac{\alpha_{k-1}^1+\alpha_0^2}{n}\right) \geq n\left(1-\frac{\varepsilon}{4}\right). \tag{5.67}$$

Therefore, for n large enough, we get the following lower bound for $n_k^1 p_k^1$:

$$n_k^1 p_k^1 \geq \frac{\beta_{k-1}^1\lambda}{n} n\left(1-\frac{\varepsilon}{4}\right)^2 \geq \left(1-\frac{\varepsilon}{2}\right)\beta_{k-1}^1\lambda, \tag{5.68}$$

where we have used once again Bernoulli's inequality in the last inequality. For the upper bound, we simply observe that using relation (5.62) yields:

$$n_k^1 p_k^1 \leq n\frac{\beta_{k-1}^1\lambda}{n} = \beta_{k-1}^1\lambda. \tag{5.69}$$

Therefore, coupling the above we get:

$$\left(1-\frac{\varepsilon}{2}\right)\beta_{k-1}^1\lambda \leq n_k^1 p_k^1 \leq \beta_{k-1}^1\lambda. \tag{5.70}$$

Now, since we have that $\underline{\beta_k^1} \leq |\partial B_{r+k}^\dagger(\rho_1)| \leq \overline{\beta_k^1}$ we conclude that $\left||\partial B_{r+k}^\dagger(\rho_1)| - n_k^1 p_k^1\right| \leq \frac{\varepsilon}{2} n_k^1 p_k^1$. Thus considering Lemma 5.2.5 with $\delta = \frac{\varepsilon}{2}$ and using the lower bound in (5.70) we get:

$$\begin{aligned}
\mathbb{P}_{\mathbf{B}_r^i}\left((\Gamma_{r,k}^1)^c \middle| \Gamma_{r,[k-1]}^1\right) &\leq \mathbb{P}_{\mathbf{B}_r^i}\left(\left||\partial B_{r+k}^\dagger(\rho_1)| - n_k^1 p_k^1\right| \geq \frac{\varepsilon}{2} n_k^1 p_k^1 \middle| \Gamma_{r,[k-1]}^1\right) \\
&\leq 2\exp\left[-\frac{(\varepsilon/2)^2 n_k^1 p_k^1}{2(1+\varepsilon/6)}\right] \\
&\leq 2\exp\left[-\frac{\varepsilon^2(1-\varepsilon/2)\beta_{k-1}^1\lambda}{8(1+\varepsilon/6)}\right] = 2e^{-q\beta_{k-1}^1\lambda},
\end{aligned} \tag{5.71}$$

where we define $q = \frac{\varepsilon^2(1-\varepsilon/2)}{8(1+\varepsilon/6)} > 0$.
We finally conclude that, using (5.61) and the fact that $\underline{\beta_{k-1}^1} \leq \beta_{k-1}^1$, our key estimate for the neighbourhood growth of an Erdős-Rényi random graph can be simplified as:

$$\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma_{r,[k]}^1\right) \geq 1-\sum_{u\in[k]} \mathbb{P}_{\mathbf{B}_r^i}\left((\Gamma_{r,u}^1)^c \middle| \Gamma_{r,[u-1]}^1\right) \geq 1-2\sum_{u\in[k]} e^{-q\beta_{k-1}^1\lambda} \geq 1-2\sum_{u\in[k]} e^{-q\underline{\beta_{k-1}^1}\lambda}. \tag{5.72}$$

Step 3: Growth of $\partial B_r(\rho_2)$

The neighbourhood growth around ρ_2 is similar to the one around ρ_1. Hence we will not give the full details on how to grow the neighbourhood of ρ_2 as the results can be adapted from Step two, where

the derivations and arguments are all very similar.
We again condition the growth on $|\partial B_{r+k}(\rho_1)| \in \left[\underline{\beta_k^1}, \overline{\beta_k^1}\right]$ and $|B_{r+k}(\rho_1)| = \alpha_k^1 \leq \overline{\alpha_k^1}$ for all set $[k]$. Similarly to the previous step, we will grow $\partial B_{r+k}(\rho_2)$ outside of $B^{\dagger}_{r+k_n^*}(\rho_1)$, and write $|\partial B^{\dagger}_{r+k}(\rho_2)|$ to denote the size of the resulting neighbourhood. In the above, we still consider $k \leq k_n^* = \lceil a \log_{\lambda(1-\varepsilon)}(n) \rceil$. We similarly define $\overline{\beta_k^2} = \beta_0^2[\lambda(1+\varepsilon)]^k$ and $\underline{\beta_k^2} = \beta_0^2[\lambda(1-\varepsilon)]^k$ as upper and lower bounds on $|\partial B^{\dagger}_{r+k}(\rho_2)|$. Hence we find a similar result as in (5.55), where:

$$\overline{\alpha_{k-1}^2} = \alpha_0^2 + \alpha_k^1 + \sum_{i=0}^{k-1} \overline{\beta_k^2} \leq C n^{a \frac{\log(\lambda(1+\varepsilon))}{\log(\lambda(1-\varepsilon))}}. \tag{5.73}$$

Furthermore, we define the following event as in (5.58):

$$\Gamma_{r,k}^2 = \left\{ \underline{\beta_k^2} \leq |\partial B^{\dagger}_{r+k}(\rho_2)| \leq \overline{\beta_k^2} \right\}, \tag{5.74}$$

where we aim to study a similar intersection as in (5.59):

$$\Gamma_{r,[k]}^2 = \bigcap_{u \in [k]} \Gamma_{r,u}^2 = \bigcap_{u \in [k]} \left\{ \underline{\beta_u^2} \leq |\partial B^{\dagger}_{r+u}(\rho_2)| \leq \overline{\beta_u^2} \right\}. \tag{5.75}$$

Now conditionally on $\Gamma_{r,k}^2$, and on $|\partial B^{\dagger}_{r+k-1}(\rho_2)| = \beta_{k-1}^2 \in \left[\underline{\beta_{k-1}^2}, \overline{\beta_{k-1}^2}\right]$ and $|B^{\dagger}_{r+k-1}(\rho_2)| = \alpha_{k-1}^2 \leq \overline{\alpha_{k-1}^2}$ we have the following aim result on the distribution of the size of $\partial B^{\dagger}_{r+k}(\rho_2)$:

$$|\partial B^{\dagger}_{r+k}(\rho_2)| \sim Bin(n_k^2, p_k^2). \tag{5.76}$$

In this case, the coefficients of the Binomial random variable are:

$$n_k^2 = n - \alpha_{k_n^*}^1 - \alpha_{k-1}^2 \qquad \text{and,} \qquad p_k^2 = 1 - \left(1 - \frac{\lambda}{n}\right)^{\beta_{k-1}^2}. \tag{5.77}$$

To again simplify the notation, we write $\mathbb{P}_{\mathbf{B}_r^i, k_n^*}$ to be the probability under the conditioning $\mathbf{B}_r^i$ and $\Gamma^2_{r,[k_n^*]}$. Then, following the same reasoning as in the previous step, we conclude a similar bound on the probability of $\Gamma^2_{r,[k_n^*]}$:

$$\mathbb{P}_{\mathbf{B}_r^i, k_n^*}\left(\Gamma^2_{r,[k_n^*]}\right) \geq 1 - 2 \sum_{u \in [k_n^*]} e^{-q\beta_{k-1}^2 \lambda} \geq 1 - 2 \sum_{u \in [k_n^*]} e^{-q\underline{\beta_{k-1}^2} \lambda}, \tag{5.78}$$

where we still consider $q = \frac{\varepsilon^2(1-\varepsilon/2)}{8(1+\varepsilon/6)} > 0$.

Step 4: Completion of the proof

Recall (5.49) where we need to find a radius $r = r(\varepsilon)$ such that for every $\varepsilon > 0$ and for every $\beta_0^1, \beta_0^2 \geq r$ and α_0^1, α_0^2 fixed,

$$\mathbb{P}(X_{n,r} | \mathbf{B}_r^i) = o(1) \Longrightarrow \limsup_{n \to +\infty} \mathbb{P}_{\mathbf{B}_r^i}(X_{n,r}) \leq \varepsilon. \tag{5.79}$$

Recall also that for $i \in \{1,2\}$, $\underline{\beta_k^i} = \beta_0^i[\lambda(1-\varepsilon)]^k$, where $\beta_0^i \geq r$. Then, if we fix $k = k_n^* = \lceil a \log_{\lambda(1-\varepsilon)}(n) \rceil$, we may take $r \to +\infty$ such that:

$$\sum_{i=1}^{k} \left(e^{-q\underline{\beta_{k-1}^1}\lambda} + e^{-q\underline{\beta_{k-1}^2}\lambda} \right) \leq \frac{\varepsilon}{2}. \tag{5.80}$$

Now, to complete the proof, it remains to study the event $\Gamma_{r,[k_n^*]} = \Gamma^1_{r,[k_n^*]} \cap \Gamma^2_{r,[k_n^*]}$. Then using the results in Step two and three along with (5.80) we have that:

$$\begin{aligned}
\liminf_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma_{r,[k_n^*]}\right) &= \liminf_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,[k_n^*]} \cap \Gamma^2_{r,[k_n^*]}\right) \\
&= \liminf_{n\to+\infty} \left[\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,[k_n^*]}\right) + \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^2_{r,[k_n^*]}\right) - \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,[k_n^*]} \cup \Gamma^2_{r,[k_n^*]}\right) \right] \\
&\geq \liminf_{n\to+\infty} \left[\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^1_{r,[k_n^*]}\right) + \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^2_{r,[k_n^*]}\right) - 1 \right] \\
&\geq \liminf_{n\to+\infty} \left[1 - 2\sum_{u\in[k_n^*]} e^{-q\underline{\beta_{k-1}^1}\lambda} + 1 - 2\sum_{u\in[k_n^*]} e^{-q\underline{\beta_{k-1}^2}\lambda} - 1 \right] \\
&= \liminf_{n\to+\infty} \left[1 - 2\sum_{i=1}^{k} \left(e^{-q\underline{\beta_{k-1}^1}\lambda} + e^{-q\underline{\beta_{k-1}^2}\lambda} \right) \right] \geq 1 - \varepsilon.
\end{aligned} \tag{5.81}$$

Hence, rewriting the above with the complement of $\Gamma_{r,[k_n^*]}$ we get:

$$\liminf_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma_{r,[k_n^*]}\right) = -\limsup_{n\to+\infty} -\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma_{r,[k_n^*]}\right) = -\limsup_{n\to+\infty} \left(\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^c_{r,[k_n^*]}\right) - 1 \right) = 1 - \limsup_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^c_{r,[k_n^*]}\right). \tag{5.82}$$

And since $\liminf_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma_{r,[k_n^*]}\right) \geq 1-\varepsilon$, it implies that $\limsup_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^c_{r,[k_n^*]}\right) \leq \varepsilon$.
Now, conditionally on $\Gamma_{r_\varepsilon,[k_n^*]}$, we use the change of base for the logarithm as in (5.54) and the fact that $\beta_0^1 \geq r = r_\varepsilon$ to have that:

$$\begin{aligned}
|\partial B_{r_\varepsilon + k_n^*}(\rho_1)| \geq \underline{\beta^1_{k_n^*}} = \beta_0^1[\lambda(1-\varepsilon)]^{k_n^*} &= \beta_0^1[\lambda(1-\varepsilon)]^{\log_{\lambda(1-\varepsilon)}(n^a)} \\
&= \beta_0^1[\lambda(1-\varepsilon)]^{\frac{\log(n^a)}{\log(\lambda(1-\varepsilon))}} \\
&= \beta_0^1 n^a \geq r_\varepsilon n^a.
\end{aligned} \tag{5.83}$$

And similarly for the neighbourhood around ρ_2:

$$|\partial B^\dagger_{r_\varepsilon + k_n^*}(\rho_2)| \geq \underline{\beta^2_{k_n^*}} = \beta_0^2[\lambda(1-\varepsilon)]^{k_n^*} = \beta_0^2 n^a \geq r_\varepsilon n^a, \tag{5.84}$$

where we still choose $a \in (\frac{1}{2}, 1)$. Hence, the total number of edges in between $\partial B_{r_\varepsilon + k_n^*}(\rho_1)$ and $\partial B^\dagger_{r_\varepsilon + k_n^*}(\rho_2)$ is at least $(r_\varepsilon n^a)^2 \gg n$. It holds that each of these potential edges may appear independently with probability $\frac{\lambda}{n}$.
Therefore, for $ER_n(\frac{\lambda}{n})$ with n vertices, we are led to consider $\mathrm{dist}_{ER_n(\frac{\lambda}{n})}(\rho_1, \rho_2)$ to be the distance between the two random roots. Then, under the given conditioning, we are interested to know if we

can reach one root ρ_1 to another root ρ_2 in less than $[2(k_n^* + r_\varepsilon) + 1]$ steps. Hence for $n \to +\infty$:

$$\begin{aligned} &\mathbb{P}_{\mathbf{B}_r^i}\left(\text{dist}_{ER_n(\frac{\lambda}{n})}(\rho_1,\rho_2) > 2(k_n^* + r_\varepsilon) + 1 \middle| \Gamma_{r_\varepsilon,[k_n^*]}\right) \leq \left(1 - \frac{\lambda}{n}\right)^{(r_\varepsilon n^a)^2} = o(1) \\ &\implies \mathbb{P}_{\mathbf{B}_r^i}\left(\text{dist}_{ER_n(\frac{\lambda}{n})}(\rho_1,\rho_2) \leq 2(k_n^* + r_\varepsilon) + 1 \middle| \Gamma_{r_\varepsilon,[k_n^*]}\right) = 1 - o(1). \end{aligned} \tag{5.85}$$

Thus, considering the Law of total probability along with (5.82) and (5.85) we get the required result (5.49) for any $\varepsilon > 0$:

$$\begin{aligned} \limsup_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}(X_{n,r}) &\leq \limsup_{n\to+\infty} \left(\mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^c_{r,[k_n^*]}\right) + \mathbb{P}_{\mathbf{B}_r^i}\left(X_{n,r} | \Gamma_{r_\varepsilon,[k_n^*]}\right)\right) \\ &\leq \limsup_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}\left(\Gamma^c_{r,[k_n^*]}\right) + \limsup_{n\to+\infty} \mathbb{P}_{\mathbf{B}_r^i}\left(X_{n,r} | \Gamma_{r_\varepsilon,[k_n^*]}\right) \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned} \tag{5.86}$$

This proves condition (5.39), where along with Theorem 5.1.1 and Theorem 5.2.2 we get the required results:

$$\frac{|\mathbf{C}_{max}|}{n} \xrightarrow{\mathbb{P}} \chi_\lambda \qquad \text{and} \qquad \frac{|\mathbf{C}_2|}{n} \xrightarrow{\mathbb{P}} 0. \tag{5.87}$$

□

Chapter 6

Conclusion

Throughout this project, the objective was always to find simpler structures to better understand (possibly large) sparse random graphs and some of their characteristics. Having introduced the notion of branching processes in the very first chapter as a relatively simple and straightforward way to study the evolution of a population in time, we understood that they could be good candidates for simplifying locally tree-like graphs. We then presented the main notion of this paper with the Benjamini-Schramm limit and went on to extend the definitions with two criteria that turned out to be more convenient to work with in general proofs. We were then able to understand the power of local limits by studying the Configuration Model where we first introduced some regularity conditions for the degrees of the vertices and by engaging in a thorough analysis of the simplicity of the model. As it turned out, we were able to understand this power by showing that the local limit of the Configuration Model is a unimodular branching process through a combinatorial argumentation.

To show how we could use local limits for the study of properties of random graphs, we applied the theory presented in the previous chapters to the analysis of the largest connected component in the graph, known as the giant component. Taking the local limit as an assumption, we were able to show that the proportion of vertices in the giant component converged to the survival probability of the associated limit, and that the proportion of vertices in the second largest component converged to zero. Given the fact that the giant component can be seen as an almost local property of the graph, we were able to end this paper by proving the previous results in the case of an Erdős-Rényi random graph under a given condition on the boundary of the neighbourhoods around two random roots.

As possible extensions of the notion of local weak convergence in a random graph setting, and continuing on a purely theoretical point of view, one might be interested to study the local limit for inhomogeneous random graphs which corresponds to a generalisation of the classical Erdős-Rényi random graphs. It was first introduced by Söderberg in [20] and extensively studied by Bollobás, Janson and Riordan in [21], where one will be confronted with the study of Poisson multiple branching processes as the associated limit. We now consider two possible applications, one in Statistical Mechanics and the other in Information Theory for Data Science. Considering the Ferromagnetic Ising Model, it was shown by Dembo and Montanari in [22] that if the model is defined on locally

tree-like graphs, then there exists a limit $\phi(\beta,B)$ to the free energy $\phi_n(\beta,B) = \frac{1}{n}\log Z_n(\beta,B)$, where they consider $B \in \mathbb{R}$ to be the magnetic field, $\beta \geq 0$ the inverse temperature and $Z_n(\beta,B)$ the partition function. As an other example, this time in Information Theory, the local limit is used as a tool to study the Stochastic Block Model which corresponds to a random graph that has the faculty to capture the effect of clustering. Such a model has been widely used as a tool for community detection (see a paper by Abbe for more details [23]), and the asymptotic behaviour of the average neighbourhood distribution for a sequence of graphs generated by a stochastic block model can be studied using the local limit, for which the associated limit is a Poisson branching process (see a paper by Alankrita Bhatt, Ziao Wang, Chi Wang and Lele Wang [24, pp.18] for a proof). Although such results are key to the understanding of complex structures in different areas of applications, one should not forget that we usually make the significant assumption of considering locally tree-like graphs. However, in a large number of cases, this simplification almost always leads to some sharp predictions of the actual models.

References

[1] P.Erdős-A.Rényi. *On the evolution of random graphs*. Publications of the Mathematical Institute of the Hungarian Academy of Sciences, 1960.

[2] E. N. Gilbert. "Random Graphs". In: *The Annals of Mathematical Statistics* 30.4 (1959), pp. 1141–1144. DOI: 10.1214/aoms/1177706098. URL: https://doi.org/10.1214/aoms/1177706098.

[3] Itai Benjamini and Oded Schramm. *Recurrence of Distributional Limits of Finite Planar Graphs*. 2001. arXiv: math/0011019 [math.PR].

[4] David Aldous and J. Steele. "The Objective Method: Probabilistic Combinatorial Optimization and Local Weak Convergence". In: *Probability on Discrete Structures* 110 (Jan. 2003). DOI: 10.1007/978-3-662-09444-0_1.

[5] Remco van der Hofstad, Pim van der Hoorn, and Neeladri Maitra. *Local limits of spatial inhomogeneous random graphs*. 2022. arXiv: 2107.08733 [math.PR].

[6] R. van der Hofstad. *Random Graphs and Complex Networks*. Vol. 1. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2017.

[7] Itai Benjamini, Russell Lyons, and Oded Schramm. "Unimodular random trees". In: *Ergodic Theory and Dynamical Systems* 35.2 (Aug. 2013), pp. 359–373. ISSN: 1469-4417. DOI: 10.1017/etds.2013.56. URL: http://dx.doi.org/10.1017/etds.2013.56.

[8] R. van der Hofstad. *Random Graphs and Complex Networks*. Vol. 2. To be published in: Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press.

[9] Lasse Leskelä. *Lecture Notes in Random Graphs and Network Statistics*. Aalto University. 2019. URL: http://math.aalto.fi/~lleskela/papers/Leskela_2019-03-27_Random_graphs_and_network_statistics.pdf.

[10] Valentas Kurauskas. *On local weak limit and subgraph counts for sparse random graphs*. 2023. arXiv: 1504.08103 [math.PR].

[11] Omer Angel, Remco van der Hofstad, and Cecilia Holmgren. *Limit laws for self-loops and multiple edges in the configuration model*. 2017. arXiv: 1603.07172 [math.PR].

[12] Universität Münster. *Size-biased Galton-Watson trees with a spine*. Lecture notes in Stochastic Processes. URL: https://www.uni-muenster.de/Stochastik/lehre/WS1011/SpezielleStochastischeProzesse/Ch_5.pdf.

[13] Remco van der Hofstad. *The giant in random graphs is almost local*. 2023. arXiv: 2103.11733v2 [math.PR].

[14] Svante Janson. *Probability asymptotics: notes on notation*. 2011. arXiv: 1108.3924 [math.PR].

[15] Alistair Sinclair. *Lecture Notes on Randomness and Computation*. Department of Computer Science, University of California Berkeley. 2022. URL: https://people.eecs.berkeley.edu/~sinclair/cs271/n16.pdf.

[16] O. Hryniv. *Lecture Notes on Stochastic Processes*. Department of Mathematics, Durham University. 2011. URL: https://www.maths.dur.ac.uk/stats/courses/StochProc34/1516RandomGraphs34.pdf.

[17] Svante Janson, Tomasz Luczak, and Andrzej Rucinski. *Random Graphs*. Wiley Series in Discrete Mathematics and Optimization. Vol. 45. Wiley, 2000. URL: https://faculty.sdu.edu.cn/_tsf/00/21/YvaIryj6ZvQf.pdf.

[18] Shanhe Wu. "Some results on extending and sharpening the Weierstrass product inequalities". In: *Journal of Mathematical Analysis and Applications* 308.2 (2005), pp. 689–702. ISSN: 0022-247X. DOI: https://doi.org/10.1016/j.jmaa.2004.11.064. URL: https://www.sciencedirect.com/science/article/pii/S0022247X04010157.

[19] Laura De Carli and Hudson Steve. "A generalization of Bernoulli's inequality". In: *Le Matematiche* 65.1 (Dec. 2010), pp. 109–117.

[20] Bo Söderberg. "General formalism for inhomogeneous random graphs". In: *Physical Review E* 66.6 (Dec. 2002). ISSN: 1095-3787. DOI: 10.1103/physreve.66.066121. URL: http://dx.doi.org/10.1103/PhysRevE.66.066121.

[21] Béla Bollobás, Svante Janson, and Oliver Riordan. "The phase transition in inhomogeneous random graphs". In: *Random Structures & amp; Algorithms* 31.1 (June 2007), pp. 3–122. ISSN: 1098-2418. DOI: 10.1002/rsa.20168. URL: http://dx.doi.org/10.1002/rsa.20168.

[22] Amir Dembo and Andrea Montanari. "Ising models on locally tree-like graphs". In: *The Annals of Applied Probability* 20.2 (Apr. 2010). ISSN: 1050-5164. DOI: 10.1214/09-aap627. URL: http://dx.doi.org/10.1214/09-AAP627.

[23] Emmanuel Abbe. *Community Detection and Stochastic Block Models*. 2023. arXiv: 1703.10146 [math.PR].

[24] Alankrita Bhatt et al. *Universal Graph Compression: Stochastic Block Models*. 2021. arXiv: 2006.02643 [cs.IT].

6CCM345a (Third Year Project) Mark Sheet

Student: Arthur Aimone
Project Title: Local Limits of Random Graphs
Supervisor: Dominic Yeo
Second Examiner: Guillaume Conchon-Kerjan

Agreed Overall Mark: 87

Comments on how the agreed mark was arrived at:

The second examiner (GCK) was agreeable to a mark in the upper half of our range based on the first examiner's report of the candidate's independence in producing this impressive thesis.

Agreed feedback comments for the student:

Overall, this was an outstanding dissertation for BSc level. The candidate exhibits a strong coverage of key ideas in the topic of local limits of sequences of graphs, as well as some other themes in random graphs more generally. The dissertation demonstrates fluency with a range of sophisticated techniques in discrete probability.

Both examiners agreed that the dissertation was very well-written and professionally presented, and indicate an author with a high personal investment in the subject. A wide range of sources have been consulted and digested in detail, then presented with an appropriate level of detail and with good style. The diagrams were valuable and showed considerable effort. Numerous comments show a deep reflection on the reasoning (as well as a good understanding of the bibliography and history), and would also probably help a non-expert reader getting into the field.

The coverage of the dissertation is very comprehensive, including theoretical aspects of local limits (in Section 5.1) and in a particular example (in Section 4.4), and background on random graphs and branching processes more generally. The thesis concluded with a good account of a highly-involved technical calculation applied to the Erdos-Renyi random graph. In other places, appropriate choices regarding when to be concise (eg convexity details in proof of Thm 2.1.2) and when to omit details (eg unimodularity for limit trees on page 10) were made. The resulting dissertation was coherent and self-contained.

If the candidate wishes to make some (minor) corrections, we noted the following:
i) the second statement of Thm 4.3.1 is incorrect (it should be \nu^2/4, as in Thm 4.3.3) and this factor of 1/2 is lost in the conversion from i<j to i =/= j during the proof. Some equalities are actually (asymptotically tight) inequalities.
ii) Around (5.9), this range of i depends on n, so to use the summation result above, need to focus on C_{<=k} instead and take the complement at the end.
iii) Around corollary 5.1.3 and Lemma 5.1.5, check for confusion between Sum (C_i)^2 and (Sum C_i)^2 in places
iv) k_n goes to infinity with n in section 5.2.3 so there should be a few more checks in the computations.

Supervisor's Marksheet

Originality and Independence: **91**
Mastery and Understanding: **87**
Extent to which the Project is Dealt With (Coverage): **80**
Presentation and Logical Structure: **92**

Overall Mark: **88**

Comments:

Overall, this was an outstanding dissertation for BSc level. The candidate exhibits a strong coverage of key ideas in the topic of local limits of sequences of graphs, as well as some other themes in random graphs more generally. The dissertation demonstrates fluency with techniques in discrete probability that go far beyond the syllabus covered at King's. Parts of the dissertation follow the canonical (graduate-level, two-volume) text by van der Hofstad. I view this as reasonable; the total length of this work is ~1000 pages, and the candidate has been selective and cited appropriately, while also discussing other sources, including original papers. Some of the chosen sources are recent, and unfamiliar to me. I am also satisfied that in some proofs and explanations the candidate has given extra detail, structure or intuition for the steps, showing full understanding. The candidate's study of the local limit for the configuration model with degrees 1,3 is successful and would be a useful example to another student meeting the topic for the first time, more so than the generalised examples given in the source texts. The candidate has also constructed a number of helpful and detailed figures of his own initiative that enhance some of the more complex arguments. As an example, page 13 contains a well-written independent explanation of the distinction between the two main forms of local weak limit and a simple example (not from the

main sources as far as I'm aware) of a graph family that satisfies one but not the other. The whole of Section 3 is successful, and probably more valuable to an undergraduate reader than any of the cited texts. With the exception of 5.2.3 (discussed shortly), the coverage of the dissertation is excellent, and the selection of material is apposite. The candidate has given a full introduction to local limits, and studied them in more detail both in a theoretical setting (5.1) and in a particular example (4.4). Sufficient background on random graphs and branching processes has been included for the dissertation to be self-contained. Appropriate choices regarding when to be concise (eg convexity details in proof of Thm 2.1.2) and when to omit details (eg unimodularity for limit trees on page 10). The mathematical writing is lucid and sophisticated, and the balance between technical steps and clarifying explanation is successful. The structure is well-considered and the candidate is clear both at chapter-by-chapter level and at section-by-section level what is being studied. The notation follows standard conventions, and is cleanly introduced. The occasional figures (eg for the configuration model) are helpful. Overall, for BSc level, the standard of presentation and mathematical exposition is very high. The only notable drawback is the length of Section 5.2.3. While this is a technically-impressive treatment of a demanding calculation argument, the connection to the rest of the work is limited. As a BSc dissertation, the overall coherence might be enhanced by ending on page 42, after the summary and figures. This might have enabled a broader introduction to random graphs, and would bring the document in line with the suggested wordcount. I have made a moderate deduction on coverage and a mild deduction on overall mark to account for this. The candidate attended regular supervision meetings, but worked mostly independently beyond the initial scope of the project, and the suggestion to study the configuration model with a particular degree distribution. My only significant intervention as supervisor was to correct one calculation involving size-biasing (where the candidate had realised that an error was present). I note that the candidate was organised and hard-working from the first week of the project, and had full drafts of many parts of the dissertation completed several months early. Thus he was able to set the agenda for all meetings, normally bringing a small, detailed set of point to discuss.

Second Examiner's Marksheet

Mastery and Understanding: **80**
Extent to which the Project is Dealt With (Coverage): **90**
Presentation and Logical Structure: **85**

Overall Mark: **83**

Comments:

Summary: This is a strong thesis, which has obviously benefited from a consequent personal investment from the student. There are only a few minor mathematical mistakes. At Bachelor level, this is overall very promising. Mastery and understanding: The student has manifestly understood a number of sophisticated concepts in probability theory, covering several related areas. There are a few minor (but non negligible) misunderstandings/mistakes in computations (at Theorem 4.3.1, Corollary 5.1.3, Lemma 5.1.5 and in the proof of Theorem 5.2.2). Numerous comments show a deep reflection on the reasoning (as well as a good understanding of the bibliography and historiography), and would also probably help a non-expert reader getting into the field. Extent: The breadth of this manuscript is remarkable for Bachelor level. It seems also that in the process, *several* references were read and understood in detail. The student investigated thoroughly a telling example of his own in Chapter 4. Presentation: The report is easy to read, there are relatively few typos. Each part starts with a clear plan for the reader (section 5.2 is a bit less obvious to follow). There are few typos overall. The pictures are well executed.